Appendix A. List of generated Conjectures

In this section, we list all the generated conjectures. We maintain a continuous interaction with HypothesiX, through which it proposes novel definitions, and we explore various properties of the defined functions to better understand their behaviour, rather than generating conjectures in unrelated directions across separate conversations. Our primary objective is to assess whether the system can meaningfully contribute to mathematical research, rather than merely suggesting simple problems.

A.1. **Generated Inequalities.**

A.1.1. *Conversation 1.*

Definition A.1. For a square-free Q where $6 \mid Q$, let U_Q be the set of residues $r \pmod Q$ such that both r and $r+2$ are coprime to Q. The residue-pairing bound is:

$$B_Q(x) = \sum_{r \in U_Q} \min\left(\pi(x; Q, r), \pi(x; Q, r+2)\right)$$

where $\pi(x; Q, a)$ is the count of primes $p \leq x$ with $p \equiv a \pmod Q$.

Conjecture A.1. For all $x \geq 7$ and any squarefree Q with $6 \mid Q$:

$$\pi_2(x) \leq B_Q(x) + 2.$$

Conjecture A.2. Explicit Modulo 30 InstanceFor $Q = 30$, $U_{30} = \{11, 17, 29\}$. Thus, for $x \geq 7$:

$$\pi_2(x) \leq 2+\min\{\pi(x; 30, 11), \pi(x; 30, 13)\}+\min\{\pi(x; 30, 17), \pi(x; 30, 19)\}+\min\{\pi(x; 30, 29), \pi(x; 30, 1)\}.$$

Conjecture A.3 (Monotone refinement)**.** If $Q_1 \mid Q_2$ are both squarefree and divisible by 6, then for every $x \geq 7$:

$$B_{Q_2}(x) \leq B_{Q_1}(x)$$

This implies that as we refine the modular partition, the sum of modular minima is non-increasing.

Definition A.2. Writing $a_Q(r; x) := \pi(x; Q, r)$ for $r \in U_Q$, one has the exact identity:

$$B_Q(x) = \frac{1}{2} \sum_{r \in U_Q} \left(a_Q(r; x) + a_Q(r+2; x) - |a_Q(r; x) - a_Q(r+2; x)|\right)$$

Conjecture A.4. Another computationally sharp inequality is:

$$\pi_2(x) \ll \frac{1}{2} S_Q(x) - \frac{1}{2} V_Q(x)$$

where $S_Q(x) := \sum_{r \in U_Q} (a_Q(r; x) + a_Q(r+2; x))$ is the total mass in admissible classes and $V_Q(x) := \sum_{r \in U_Q} |a_Q(r; x) - a_Q(r+2; x)|$ is the summed class-discrepancy.

Conjecture A.5. A coarser π-only corollary replaces $\frac{1}{2}(a_Q(r; x)+a_Q(r+2; x))$ by $\min(a_Q(r; x), a_Q(r+2; x)) \leq \frac{1}{2}(a_Q(r; x) + a_Q(r+2; x))$, eliminating explicit absolute values:

$$\pi_2(x) \leq 2 + \frac{1}{2}\left(\pi(x) - \sum_{a \in F_Q} \pi(x; Q, a) - c_Q\right)$$

where $F_Q := \{a \pmod Q : \gcd(a, Q) = 1 \text{ but } \gcd(a+2, Q) \neq 1\}$ are forbidden start-classes and c_Q removes the small primes dividing Q (explicit below per modulus).

Definition A.3. For $Q_1 \mid Q_2$, let the fiber $F_{Q_2 \to Q_1}(r_1) := \{r_2 \in U_{Q_2} : r_2 \equiv r_1 \pmod{Q_1}\}$.

Conjecture A.6. If $Q_1 \mid Q_2$ are squarefree with $6 \mid Q_1$, then for all $x \geq 7$:

$$B_{Q_2}(x) \leq \sum_{r_1 \in U_{Q_1}} \min \left(\sum_{r_2 \in F} a_{Q_2}(r_2; x), \sum_{r_2 \in F} a_{Q_2}(r_2 + 2; x) \right) \leq B_{Q_1}(x).$$

Conjecture A.7 (Exact fiber cancellation identity (Gap decomposition))**.** For each $r_1 \in U_{Q_1}$, let $L(r_1; x)$ and $R(r_1; x)$ be the sums of the left and right counts in the fiber. The fiber's cancellation deficit is:

$$\delta_{r_1}(x) := \frac{1}{2} \left[\sum_{r_2 \in F} |d_{r_2}(x)| - \left| \sum_{r_2 \in F} d_{r_2}(x) \right| \right] \geq 0$$

where $d_{r_2}(x) = a_{Q_2}(r_2; x) - a_{Q_2}(r_2 + 2; x)$. The total gap splits as:

$$B_{Q_1}(x) - B_{Q_2}(x) = \Phi_{Q_1 \leftarrow Q_2}(x) + \sum_{r_1 \in U_{Q_1}} \delta_{r_1}(x)$$

where Φ is the loss from residue classes that become forbidden modulo Q_2.

Conjecture A.8 (Strict Drop)**.** Fix squarefree Q and prime $p \geq 5$ coprime to Q. There exists X_0 such that for all $x \geq X_0$:

$$B_{pQ}(x) \leq B_Q(x) - 1 \quad \text{(SD)}$$

This predicts a persistent strict drop along primorial refinement due to the emergence of forbidden residue families and fiber imbalances.

Definition A.4 (general h-step tuples)**.** We define:

- Let $h \in 2\mathbb{N}$ and $m \geq 1$. An h-step $(m+1)$-tuple is a set of primes $\{p, p+h, \ldots, p+mh\}$.
- For a modulus $Q \geq 2$ define the admissible starts $U_Q^{(h,m)} := \{r \pmod{Q} : \gcd(r + jh, Q) = 1 \text{ for all } j = 0, \ldots, m\}$.
- For a residue-class prime count $a_Q(a; x) := \pi(x; Q, a)$ and a tuple-minimum over a fixed start r:
$$M_Q^{(h,m)}(r; x) := \min_{0 \leq j \leq m} a_Q(r + jh; x).$$
- Define the modular minima bound:
$$B_Q^{(h,m)}(x) := \sum_{r \in U_Q^{(h,m)}} M_Q^{(h,m)}(r; x).$$

Conjecture A.9. For all $x \geq 7$:

$$N_{h,m}(x) \leq C_{h,m}(Q) + B_Q^{(h,m)}(x) \quad (*)$$

where $N_{h,m}(x)$ counts h-step $(m+1)$-tuples with $p + mh \leq x$, and $C_{h,m}(Q)$ is a small explicit constant accounting for tuples involving small primes dividing Q.

Conjecture A.10 (Example)**.** Cousin primes (h=4), m=1
Q=6: admissible start $r \equiv 1 \pmod 6$; end $r + 4 \equiv 5 \pmod 6$. For $x \geq 7$:

$$\pi_4(x) \leq 1 + \min\{\pi(x; 6, 1), \pi(x; 6, 5)\}$$

The constant 1 accounts for the exceptional pair $(3, 7)$.

A.1.2. *Conversation 2.*

Definition A.5. Let

- $\pi_2(x) := \#\{p \le x-2 : p \text{ and } p+2 \text{ are prime}\}$.
- $\pi_4(x) := \#\{p \le x-4 : p \text{ and } p+4 \text{ are prime}\}$.
- $\pi(x;q,a)$ is the count of primes $\le x$ that are $\equiv a \pmod q$.

Conjecture A.11 (Structural inequality)**.** For all $x \ge 7$:

$$\pi_2(x) + \pi_4(x) \le \pi(x-4) + 2$$

This follows from the impossibility of prime triplets $\{p, p+2, p+4\}$ beyond $\{3,5,7\}$.

Conjecture A.12 (Twin-cousin bias inequality)**.** There exist absolute constants $c_1, c_2 > 0$ and x_0 such that for all $x \ge x_0$:

$$c_1 \cdot \frac{\pi(x;3,2) - \pi(x;3,1)}{\log x} - \frac{x}{(\log x)^3} \le \pi_2(x) - \pi_4(x) \le c_2 \cdot \frac{\pi(x;3,2) - \pi(x;3,1)}{\log x} + \frac{x}{(\log x)^3}.$$

In particular, the set of x for which $\pi_2(x) > \pi_4(x)$ has logarithmic density strictly larger than $1/2$.

Definition A.6. Let

- For even $h \ge 2$, write $\pi_h(x) := \#\{p \le x-h : p \text{ and } p+h \text{ are prime}\}$.
- For modulus $q \ge 1$ and residue a modulo q, let $\pi(x;q,a) := \#\{p \le x : p \text{ prime}, p \equiv a \pmod q\}$ and $\Delta_q(a;x) := \pi(x;q,a) - \pi(x)/\phi(q)$.
- Define the admissibility mask $K_{q,h}(a) := 1_{\gcd(a(a+h),q)=1}$ (i.e., a and $a+h$ avoid all prime divisors of q).
- The mod-q pair-bias transform for gap h is:

  $$B_{q,h}(x) := \frac{1}{\log x} \sum_{a \pmod q} K_{q,h}(a) \cdot \Delta_q(a;x)$$

  and for two even gaps h_1, h_2 set:

  $$R_{q;h_1,h_2}(x) := B_{q,h_1}(x) - B_{q,h_2}(x) = \frac{1}{\log x} \sum_{a \pmod q} (K_{q,h_1}(a) - K_{q,h_2}(a)) \cdot \Delta_q(a;x).$$

Conjecture A.13 (Structural obstruction inequality)**.** Let $r \ge 3$ be an odd prime. Then for all $x \ge 2r$:

$$\sum_{j=1}^{r-1} \pi_{2j}(x) \le (r-2) \cdot \pi(x - 2(r-1)) + O_r(1)$$

In particular, for $r = 3$ this yields $\pi_2(x) + \pi_4(x) \le \pi(x-4) + O(1)$, recovering and sharpening the classical mod-3 obstruction.

Conjecture A.14 (Universal mod-q pair-bias inequality)**.** Let $q \ge 3$ be squarefree and $h_1, h_2 \ge 2$ be even. Then there exists $C(q,h_1,h_2) > 0$ and x_0 such that for all $x \ge x_0$:

$$|\pi_{h_1}(x) - \pi_{h_2}(x) - R_{q;h_1,h_2}(x)| \le C(q,h_1,h_2) \cdot \frac{x}{(\log x)^3}.$$

Equivalently, the sign and size of $\pi_{h_1}(x) - \pi_{h_2}(x)$ at scale $\ll x/(\log x)^2$ are governed by the weighted prime-race imbalance $R_{q;h_1,h_2}(x)$.

Conjecture A.15 (Concrete Specialization)**.** For any q multiple of 6, $K_{q,2} - K_{q,4}$ is supported on classes a with $a \equiv 1$ or $5 \pmod 6$ and equals -1 on $a \equiv 1 \pmod 6$, $+1$ on $a \equiv 5 \pmod 6$. Hence:

$$R_{q;2,4}(x) = \frac{1}{\log x}(\pi(x;6,5) - \pi(x;6,1)).$$

Thus (2) predicts that $\pi_2(x) - \pi_4(x)$ tracks the classical Chebyshev bias between 5 and 1 modulo 6, uniformly across all squarefree moduli q divisible by 6.

Basic Properties (all q squarefree, even h):

(P1) Linearity: $B_{q,h}(x)$ is linear in the vector $(\Delta_q(a;x))_a$, and $R_{q;h_1,h_2}$ is linear in K-differences.

(P2) Locality/CRT: If $q = \prod p_i$ is squarefree, then $K_{q,h}(a) = \prod_i 1_{a \not\equiv 0,-h \pmod{p_i}}$ and $B_{q,h}$ factors across primes dividing q.

(P3) Stability under refinement: If $q|Q$ (both squarefree), then the pushforward of weights from modulo Q to modulo q preserves R in the sense that $\sum_{b \equiv a \pmod q}(K_{Q,h_1} - K_{Q,h_2})(b) = (K_{q,h_1} - K_{q,h_2})(a)$.

(P4) Twin–cousin universality: For any squarefree $q \equiv 0 \pmod 6$, identity (3) holds; thus the twin–cousin discrepancy depends only on the prime-race $\pi(x;6,5)$ vs. $\pi(x;6,1)$ at main bias scale.

(P5) Scale: Under standard distribution in AP on average, the error $x/(\log x)^3$ is natural for sieve-based pair counts.

Definition A.7. For $q = 6$:

- $\pi_h(x) := \#\{p \le x - h : p \text{ and } p + h \text{ are prime}\}$ for even $h \ge 0$, so $\pi_0(x) = \pi(x)$.
- $\pi(x;6,a) := \#\{p \le x \text{ prime} : p \equiv a \pmod 6\}$ and $\Delta_6(a;x) := \pi(x;6,a) - \pi(x)/\phi(6)$ with $\phi(6) = 2$.
- $\omega_6(x) := \#\{\text{primes } p \le x \text{ with } p \mid 6\} \in \{0,1,2\}$.
- $K_{6,h}(a) := 1_{\gcd(a(a+h),6)=1}$. For $h = 0,2,4$, one has $K_{6,0} = 1_{a\in\{1,5\}}$, $K_{6,2} = 1_{a=5}$, $K_{6,4} = 1_{a=1}$.
- Bias transform: $B_{6,h}(x) := (1/\log x) \cdot \sum_{a \pmod 6} K_{6,h}(a) \cdot \Delta_6(a;x)$.

Conjecture A.16 (Exact Triad Identities)**.** **(I1)** $B_{6,2}(x) = \Delta_6(5;x)/\log x$**,** $B_{6,4}(x) = \Delta_6(1;x)/\log x$**,** $B_{6,0}(x) = (\Delta_6(1;x) + \Delta_6(5;x))/\log x$.

(I2) Mass-conservation and race-duality:

$$B_{6,2}(x) + B_{6,4}(x) = B_{6,0}(x) = -\omega_6(x)/\log x$$

$$B_{6,2}(x) - B_{6,4}(x) = (\pi(x;6,5) - \pi(x;6,1))/\log x.$$

Conjecture A.17 (Elementary Start/End Class Caps)**.** For all $x \ge 7$

- (C1) Start-side caps: $\pi_2(x) \le \pi(x-2;6,5) + 1$, $\pi_4(x) \le \pi(x-4;6,1) + 1$.
- (C2) End-side caps: $\pi_2(x) \le \pi(x;6,1) + 1$, $\pi_4(x) \le \pi(x;6,5) + 1$.
- (O) Aggregate Obstruction: $\pi_2(x) + \pi_4(x) \le \pi(x-4) + 2$.

Conjecture A.18 (Conjectural Bias-Inequality)**.** There exists an absolute $C > 0$ and x_0 such that for all $x \ge x_0$:

$$|\pi_2(x) - \pi_4(x) - (\pi(x;6,5) - \pi(x;6,1))/\log x| \le C \cdot x/(\log x)^3$$

Equivalently, the twin-cousin discrepancy is governed by the explicit prime race between 5 and 1 modulo 6 at amplitude $1/\log x$, up to a sieve-scale error.

Definition A.8. Define:

- Prime Counts: For even $h \ge 0$, $\pi_h(x) := \#\{p \le x - h : p \text{ and } p + h \text{ are prime}\}$ with $\pi_0(x) = \pi(x)$.
- Arithmetic Progressions: For modulus $q \ge 1$ and residue $a \pmod q$, $\pi(x;q,a) := \#\{p \le x : p \text{ prime}, p \equiv a \pmod q\}$, $\Delta_q(a;x) := \pi(x;q,a) - \pi(x)/\phi(q)$.
- Admissibility Mask: $K_{q,h}(a) := 1_{\gcd(a(a+h),q)=1}$.
- Residuewise Bias Transform: $B_{q,h}(a;x) := (1/\log x) \cdot K_{q,h}(a) \cdot \Delta_q(a;x)$.

- Min-Bias Aggregator: For gaps 2 and 4, define

$$F_{q;2,4}(x) := \sum_{a \pmod q} \min(B_{q,2}(a;x), B_{q,4}(a;x)).$$

- Small Primes Count: $\omega_q(x) := \#\{\text{primes } p \le x \text{ with } p \mid q\}$.

Conjecture A.19. For any modulus q divisible by 6 and all $x \ge 7$:

$$F_{q;2,4}(x) = -\frac{\omega_q(x) + T_q(x)}{2 \cdot \log x}$$

where

$$T_q(x) := \sum_{a \pmod q, \gcd(a,q)=1} |\Delta_q(a;x)|$$

is the total variation of the centered prime counts across all reduced residue classes modulo q. In particular, $F_{q;2,4}(x) \le 0$ and becomes more negative as either $\omega_q(x)$ or the dispersion $T_q(x)$ increases.

Conjecture A.20 (Simple Universal Bounds and Equality Cases)**.** (B1) Two-sided bounds:

$$-\frac{|d(x)| + 2\omega_q(x)}{2 \cdot \log x} \le F_{q;2,4}(x) \le -\frac{\omega_q(x)}{\log x} \le 0$$

where $d(x) := \pi(x;6,5) - \pi(x;6,1)$. (Note: $d(x)$ is q-independent for q multiple of 6.)

(B2) Sharper sandwich ($q = 6$): With $B_{6,0}(x) := -\omega_6(x)/\log x$ and $R(x) := (\pi(x;6,5) - \pi(x;6,1))/\log x$:

$$B_{6,0}(x) - \frac{|R(x)|}{2} \le F_{6;2,4}(x) \le B_{6,0}(x) \le 0.$$

Definition A.9. For q squarefree, $q \equiv 0 \pmod 6$, $x \ge 7$:

- $\Delta_q(a;x) := \pi(x;q,a) - \pi(x)/\phi(q)$, for $(a,q) = 1$; set $\Delta_q(a;x) = 0$ otherwise.
- Masks $K_{q,2} = 1$ on reduced residues $a \equiv 5 \pmod 6$, $K_{q,4} = 1$ on $a \equiv 1 \pmod 6$, else 0.
- $B_{q,h}(x) := (1/\log x) \sum_{a \pmod q} K_{q,h}(a) \Delta_q(a;x)$.
- $\omega_q(x) := \#\{\text{primes } p \le x \text{ with } p \mid q\}$. Let $\omega_q^{(1)}(x), \omega_q^{(5)}(x)$ be the counts of prime divisors of $q \le x$ that are $\equiv 1, 5 \pmod 6$, respectively.

Conjecture A.21. Define $S_q(x) := B_{q,2}(x) + B_{q,4}(x) = -\omega_q(x)/\log x$, and

$$D_q(x) := B_{q,2}(x) - B_{q,4}(x) = \frac{\pi(x;6,5) - \pi(x;6,1) - (\omega_q^{(5)}(x) - \omega_q^{(1)}(x))}{\log x}.$$

Then $B_{q,2}(x) = (S_q(x) + D_q(x))/2$ and $B_{q,4}(x) = (S_q(x) - D_q(x))/2$.

Conjecture A.22 (Ellipse law)**.**

$$(B_{q,2}(x) - S_q(x)/2)^2 + (B_{q,4}(x) - S_q(x)/2)^2 = D_q(x)^2/2$$

In particular, the pair $(B_{q,2}, B_{q,4})$ lies on a circle of radius $|D_q|/\sqrt{2}$ centered at $(S_q/2, S_q/2)$, so $|B_{q,2} - B_{q,4}| = |D_q|$.

Conjecture A.23 (Product identity and bounds)**.**

$$4B_{q,2}(x)B_{q,4}(x) = S_q(x)^2 - D_q(x)^2.$$

Hence:

$$-D_q(x)^2/4 \le B_{q,2}(x)B_{q,4}(x) \le S_q(x)^2/4 = \frac{\omega_q(x)^2}{4\log^2 x}.$$

Writing the total variation $T_q(x) := \sum_{(a,q)=1} |\Delta_q(a;x)|$, we have $|D_q(x)| \le T_q(x)/\log x$, so:

$$\frac{S_q(x)^2 - (T_q(x)/\log x)^2}{4} \le B_{q,2}(x)B_{q,4}(x) \le S_q(x)^2/4.$$

Conjecture A.24 (L^1/L^2 control of the race)**.** Let $\chi_6(a) = +1$ on $a \equiv 5 \pmod 6$, -1 on $a \equiv 1 \pmod 6$, lifted to modulo q.

$$|D_q(x)| \le T_q(x)/\log x$$

$$|D_q(x)| \le (1/\log x)\sqrt{\phi(q) \cdot V_q(x)}, \text{ where } V_q(x) := \sum_{(a,q)=1} \Delta_q(a;x)^2.$$

Conjecture A.25 (Sum and max of absolute values)**.**

$$|B_{q,2}(x)| + |B_{q,4}(x)| \ge \max(|S_q(x)|, |D_q(x)|)$$

$$|B_{q,2}(x)| + |B_{q,4}(x)| \le \sqrt{S_q(x)^2 + D_q(x)^2}$$

$$\frac{(\big||S_q(x)| - |D_q(x)|\big|)}{2} \le \max(|B_{q,2}(x)|, |B_{q,4}(x)|) \le \frac{|S_q(x)| + |D_q(x)|}{2}.$$

Conjecture A.26 (One-sided sign constraints and bracketingSince)**.** $S_q(x) \le 0$, at least one of $B_{q,2}(x), B_{q,4}(x)$ is ≤ 0. Furthermore:

$$\min(B_{q,2}(x), B_{q,4}(x)) \le S_q(x)/2 \le \max(B_{q,2}(x), B_{q,4}(x)).$$

A.1.3. *Conversation 3.*

Conjecture A.27. For all $x \ge 7$:

$$|\pi_2(x) - \pi_4(x)| \le |\pi(x;6,5) - \pi(x;6,1)|$$

Here $\pi(x;6,a)$ counts primes $\le x$ congruent to a modulo 6. In particular, the conjecture predicts that the excess of twin over cousin pairs up to x can never exceed the Chebyshev bias between primes $\equiv 5$ and $\equiv 1 \pmod 6$.

Quantitative Refinement: (Expected under standard hypotheses such as GRH for Dirichlet L-functions modulo 6, and heuristic Hardy–Littlewood prime-pair asymptotics). There exists an absolute constant $K > 0$ such that, for all $x \ge 7$:

$$|\pi_2(x) - \pi_4(x)| \le K\frac{x^{1/2}}{\log x}.$$

Definition A.10. For even $h \ge 2$ with $3 \nmid h$, let:

$$\pi_h(x) := \#\{p \le x - h : p \text{ and } p + h \text{ are prime}\}$$

Let $r(h) \in \{1,5\}$ be the unique residue class modulo 6 such that $p \equiv r(h) \pmod 6$ forces p and $p+h$ (both > 3) to be prime-residue admissible; explicitly $r(h) = 5$ if $h \equiv 2 \pmod 6$ and $r(h) = 1$ if $h \equiv 4 \pmod 6$. Let:

$$S_h(x) := \pi(x-h;6,r(h)) = \#\{p \le x - h : p \text{ prime and } p \equiv r(h) \pmod 6\}$$

Fix a finite $H_2 \subseteq \{h \ge 2 : h \equiv 2 \pmod 6, 3 \nmid h\}$ and $H_4 \subseteq \{h \ge 2 : h \equiv 4 \pmod 6, 3 \nmid h\}$. Let nonnegative weights $(w_h)_{h \in H_2 \cup H_4}$ satisfy $\sum_{h\in H_2} w_h = \sum_{h \in H_4} w_h = 1$.

Conjecture A.28 (Start-Normalized Mod-6 Parity Inequality)**.** There exist absolute constants $x_0 \geq 7$ and $C > 0$ such that for all $x \geq x_0$:

$$\left| \sum_{h \in H_2} w_h \cdot \frac{\pi_h(x)}{S_h(x)} - \sum_{h \in H_4} w_h \cdot \frac{\pi_h(x)}{S_h(x)} \right| \leq \frac{C}{\log x} \quad (1)$$

Moreover, for each fixed finite choice of H_2, H_4 and weights, the left-hand side is $O((\log x)^{-1-\delta})$ for some $\delta = \delta(H_2, H_4, w) > 0$ under GRH for Dirichlet L-functions modulo 6 together with the Hardy–Littlewood prime-pair conjecture for all $h \in H_2 \cup H_4$.

Conjecture A.29. Special Cases and Corollaries:

- Twin vs cousin (unweighted): take $H_2 = \{2\}, H_4 = \{4\}, w_2 = w_4 = 1$, yielding:
$$\left| \frac{\pi_2(x)}{\pi(x-2; 6, 5)} - \frac{\pi_4(x)}{\pi(x-4; 6, 1)} \right| \leq \frac{C}{\log x}.$$
- Symmetric 2-gap vs 4-gap families: for any $m \geq 0$, with $H_2 = \{6j + 2 : 0 \leq j \leq m\}$, $H_4 = \{6j + 4 : 0 \leq j \leq m\}$ and uniform weights $w_h = 1/(m + 1)$:
$$\left| \frac{1}{m+1} \sum_{j=0}^{m} \frac{\pi_{6j+2}(x)}{\pi(x - (6j+2); 6, 5)} - \frac{1}{m+1} \sum_{j=0}^{m} \frac{\pi_{6j+4}(x)}{\pi(x - (6j+4); 6, 1)} \right| \leq \frac{C}{\log x}.$$
- A simpler, coarser version is $|\pi_2(x) - \pi_4(x)| \leq |\pi(x-2; 6, 5) - \pi(x-4; 6, 1)| + O(1)$.

Definition A.11. Let $q \in \mathbb{N}$ with $6 \mid q$. For even $h \geq 2$, define:

- $\pi_h(x) := \#\{p \leq x - h : p \text{ and } p + h \text{ are prime}\}$.
- $\pi(x; q, a) := \#\{p \leq x : p \text{ prime}, p \equiv a \pmod q\}$.
- $\pi_h(x; q, a) := \#\{p \leq x - h : p \equiv a \pmod q, p \text{ and } p + h \text{ prime}\}$.

Define the two mod-6 fibers inside $(\mathbb{Z}/q\mathbb{Z})^\times$ by:

- $F_5(q) := \{a \pmod q : a \equiv 5 \pmod 6 \text{ and } \gcd(a, q) = 1\}$.
- $F_1(q) := \{a \pmod q : a \equiv 1 \pmod 6 \text{ and } \gcd(a, q) = 1\}$.

Aggregate fiber counts and start supplies:

- $\Pi_2(x; q) := \sum_{a \in F_5(q)} \pi_2(x; q, a)$.
- $\Pi_4(x; q) := \sum_{a \in F_1(q)} \pi_4(x; q, a)$.
- $S_5(x; q) := \sum_{a \in F_5(q)} \pi(x - 2; q, a)$.
- $S_1(x; q) := \sum_{a \in F_1(q)} \pi(x - 4; q, a)$.

Conjecture A.30 (Fibered, Start-Normalized Twin–Cousin Near-Equality)**.** There exist constants $x_0(q)$ and $C(q) > 0$ such that for all $x \geq x_0(q)$:

$$\left| \frac{\Pi_2(x; q)}{S_5(x; q)} - \frac{\Pi_4(x; q)}{S_1(x; q)} \right| \leq \frac{C(q)}{\log x} \quad \text{(A)}$$

Heuristically, the left-hand side is expected to be $\ll_q 1/(\log x)^2$ after cancellation of the common leading term $4C_2/\log x$.

Conjecture A.31 (Coarse, Unnormalized Envelope by Fibered Chebyshev Bias)**.** For all $x \geq 7$:

$$|\Pi_2(x; q) - \Pi_4(x; q)| \leq |S_5(x; q) - S_1(x; q)| + 1 \quad \text{(B)}$$

The $+1$ accounts for the exceptional pair starting at $p = 3$.

Notation. Let q be a positive integer with $6 \mid q$. For a finite set of even gaps $\Delta \subseteq 2\mathbb{N}$ with $0 \in \Delta$, write $M(\Delta) := \max \Delta$.

- $\pi(x; q, a) := \#\{p \le x : p \text{ prime}, p \equiv a \pmod q\}$.
- $\pi_\Delta(x; q, a) := \#\{p \le x - M(\Delta) : p \equiv a \pmod q \text{ and } p + d \text{ is prime for all } d \in \Delta\}$.

Definition A.12. Define fibers $F_5(q)$ and $F_1(q)$ as residues $a \pmod q$ congruent to 5 and 1 $\pmod 6$ respectively.

Define fiber-admissible residue sets:$F_r(\Delta; q) := \{a \in F_r(q) : \gcd(a + d, q) = 1 \text{ for all } d \in \Delta\}, r \in \{1, 5\}$.

Define aggregated counts and start supplies:

- $\Pi_\Delta(x; q, r) := \sum_{a \in F_r(\Delta;q)} \pi_\Delta(x; q, a)$.
- $S_r^\Delta(x; q) := \sum_{a \in F_r(\Delta;q)} \pi(x - M(\Delta); q, a)$.

We write $A(x) \ll_{q,\Delta} B(x)$ to mean $|A(x)| \le C(q, \Delta)B(x)$ for some constant $C(q, \Delta)$.

Conjecture A.32 (Intra-family Fiber Balance for Sexy Pairs and Chains)**.** For any finite sexy-chain tuple $\Delta \subseteq 6\mathbb{N}$ with $0 \in \Delta$ (e.g., $\Delta = \{0, 6\}$), for all sufficiently large x:

$$\left| \frac{\Pi_\Delta(x; q, 1)}{S_1^\Delta(x; q)} - \frac{\Pi_\Delta(x; q, 5)}{S_5^\Delta(x; q)} \right| \ll_{q,\Delta} \frac{1}{\log x}$$

Conjecture A.33 (Cross-family Coupling Between Complementary Fiber-orientations)**.** Let Δ_2, Δ_4 be finite even-gap tuples with $3 \nmid d$ for all $d \in \Delta_2 \cup \Delta_4$ forcing starts in fibers 5 and 1 respectively. For any nonnegative weights:

$$\left| \sum_{d \in \Delta_2} w_d \cdot \frac{\Pi_{\{0,d\}}(x; q, 5)}{S_5^{\{0,d\}}(x; q)} - \sum_{e \in \Delta_4} v_e \cdot \frac{\Pi_{\{0,e\}}(x; q, 1)}{S_1^{\{0,e\}}(x; q)} \right| \ll_{q,\Delta_2,\Delta_4} \frac{1}{\log x}.$$

Conjecture A.34 (Coarse Bias Envelopes (Unnormalized))**.** For any tuple Δ with both fibers admissible:

$$|\Pi_\Delta(x; q, 1) - \Pi_\Delta(x; q, 5)| \ll_{q,\Delta} |S_1^\Delta(x; q) - S_5^\Delta(x; q)| + 1.$$

A.1.4. *Conversation 4.*

Definition A.13. Define:

- $\Pi_2(x) := \#\{p \le x : p \text{ and } p + 2 \text{ are prime}\}$ (twin-prime counting function).
- $\Pi_4(x) := \#\{p \le x : p \text{ and } p + 4 \text{ are prime}\}$ (cousin-prime counting function).
- $\pi(x; 3, a) := \#\{p \le x : p \text{ is prime and } p \equiv a \pmod 3\}$, for $a \in \{1, 2\}$.
- $\Delta_{\text{pairs}}(x, y) := (\Pi_2(x + y) - \Pi_2(x)) - (\Pi_4(x + y) - \Pi_4(x))$.
- $\Delta_3(x, y) := (\pi(x + y; 3, 2) - \pi(x; 3, 2)) - (\pi(x + y; 3, 1) - \pi(x; 3, 1))$.
- $C_2 := \prod_{p>2}(p(p - 2)/(p - 1)^2) \approx 0.6601618158$ (the twin-prime constant). Define $\lambda(x) := 2C_2/\log x$.

Conjecture A.35 (Twin-Cousin Linear Response in Short Intervals)**.** There exist absolute constants $x_0 \ge 10$, $\theta \in (0, 1)$, and $K > 0$ such that for all $x \ge x_0$ and $x^\theta \le y \le x$,

$$|\Delta_{\text{pairs}}(x, y) - \lambda(x) \cdot \Delta_3(x, y)| \le K \cdot \sqrt{y}/(\log x)^2.$$

Equivalently, a one-sided form holds:

$$|\Delta_{\text{pairs}}(x, y)| \le (2C_2/\log x) \cdot |\Delta_3(x, y)| + K \cdot \sqrt{y}/(\log x)^2.$$

Conjecture A.36 (Cumulative one-variable form)**.** For $t \ge t_0$ (sufficiently large),

$$|\Pi_2(t) - \Pi_4(t) - (2C_2/\log t) \cdot (\pi(t; 3, 2) - \pi(t; 3, 1))| \le K \cdot \sqrt{t}/(\log t)^2.$$

Open directions.

- Prove Conjecture A.35 under GEH-type hypotheses by adapting dispersion estimates.

- Optimize the exponent θ and constant K.
- Develop a semiprime proxy inequality by relating $\Delta_3(x,y)$ to a smoothed residue-class imbalance for Q in $(x,x+y]$.

A.1.5. *Conversation 5.*

Definition A.14. Write $\pi_2(x)$ for the number of twin-prime pairs $(p,p+2)$ with $p+2\le x$, and $\pi_4(x)$ for the number of cousin-prime pairs $(p,p+4)$ with $p+4\le x$. Let $Q(x)$ denote the number of semiprimes $n\le x$ ($\Omega(n)=2$, counted with multiplicity). For $x\ge x_0$ and $h\ge 1$ define the short-interval discrepancy:

$$\Delta_h(x) := (\pi_2(x+h)-\pi_2(x)) - (\pi_4(x+h)-\pi_4(x)).$$

Conjecture A.37 (self-normalized square-root law)**.** There exist absolute constants $A,B>0$ and $\theta\in(0,1)$ such that for all $x\ge x_0$ and $x^\theta\le h\le x$,

$$|\Delta_h(x)| \le A\sqrt{(\pi_2(x+h)-\pi_2(x)) + (\pi_4(x+h)-\pi_4(x))} + B.$$

Equivalently, for the normalized counts,

$$\left|\frac{\pi_2(x+h)-\pi_2(x)}{h/\log^2 x} - \frac{\pi_4(x+h)-\pi_4(x)}{h/\log^2 x}\right| \le \frac{A'}{\sqrt{h/\log^2 x}} + \frac{B'}{h/\log^2 x}$$

with absolute $A',B'>0$, whenever $x^\theta\le h\le x$.

Conjecture A.38 (semiprime control)**.** There exists an absolute constant $C>0$ and $\theta'\in(0,1)$ such that for all $x\ge x_0$ and $x^{\theta'}\le h\le x$,

$$|\Delta_h(x)| \le C\sqrt{Q(x+h)-Q(x)}.$$

Definition A.15. For $x\ge 2$ and $h\ge 1$, let $I=(x,x+h]$, and define:

- $\pi_2(I) := \#\{p\in I : p, p+2 \text{ are prime}\}$, $\pi_4(I) := \#\{p\in I : p, p+4 \text{ are prime}\}$.
- For $r\in\{1,5,7,11\}$, let $P_r(I) := \#\{p\in I : p\equiv r \pmod{12}\}$.
- Let $R(I) := \pi_2(I)/\pi_4(I)$ whenever $\pi_4(I)>0$.

Conjecture A.39 (Residue-ratio sandwich for twin vs. cousin pairs)**.** There exists $\theta\in(0,1)$ and $x_0\ge 2$ such that for all $x\ge x_0$ and $x^\theta\le h\le x$ with $\pi_4(I)>0$,

$$\min\{P_1(I)/P_7(I), P_7(I)/P_1(I)\} \le R(I) \le \max\{P_1(I)/P_7(I), P_7(I)/P_1(I)\}.$$

Conjecture A.40 (Robust form with a universal distortion)**.** There exists an absolute $\kappa\ge 1$ such that for all such I,

$$\kappa^{-1}\cdot\min\{P_1(I)/P_7(I), P_7(I)/P_1(I)\} \le R(I) \le \kappa\cdot\max\{P_1(I)/P_7(I), P_7(I)/P_1(I)\}.$$

Conjecture A.41 (Semiprime proxy variant)**.** Let $Q_r(I) := \#\{n\in I : n \text{ is semiprime and } n\equiv r \pmod{12}\}$. Then for some $\theta'\in(0,1)$, $x\ge x_0'$, $x^{\theta'}\le h\le x$ with $\pi_4(I)>0$,

$$\min\{(Q_1(I)/Q_7(I))^{1/2}, (Q_7(I)/Q_1(I))^{1/2}\} \le R(I) \le \max\{(Q_1(I)/Q_7(I))^{1/2}, (Q_7(I)/Q_1(I))^{1/2}\}.$$

Definition A.16. Fix modulus 30. Let $R=\{1,7,11,13,17,19,23,29\}$ be the admissible odd prime residues mod 30. For an interval $I=(x,x+h]$, define:

- $P_r(I) := \#\{p\in I \text{ prime} : p\equiv r \pmod{30}\}$ for $r\in R$.
- For even gaps $h\in\{2,4,6\}$, let $\pi_h(I) := \#\{p\in I \text{ prime} : p+h\in I \text{ and } p+h \text{ is prime}\}$.

Admissible residue starts and targets (mod 30):

- Twin (h=2): starts $S_2=\{11,17,29\}$, targets $T_2=\{1,13,19\}$.
- Cousin (h=4): starts $S_4=\{7,13,19\}$, targets $T_4=\{11,17,23\}$.

- Sexy (h=6): starts $S_6 = \{1, 7, 11, 13, 17, 23\}$, targets $T_6 = \{7, 13, 17, 19, 23, 29\}$.

For each h, set $A_h(I) := \sum_{r \in S_h} P_r(I)$, and define:

$$\mu_h^-(I) := \min_{s \in T_h} P_s(I), \quad \mu_h^+(I) := \max_{s \in T_h} P_s(I).$$

Define $T_h(I) := \sum_{r \in S_h} P_r(I) P_{r+h}(I)$.

Then by the elementary inequality $\min b_i \sum a_i \leq \sum a_i b_i \leq \max b_i \sum a_i$, we have:

$$\mu_h^-(I) A_h(I) \leq T_h(I) \leq \mu_h^+(I) A_h(I).$$

Conjecture A.42 (mod-30 ratio sandwiches for twin, cousin, sexy)**.** There exists a universal distortion $\kappa \geq 1$ and $\theta \in (0, 1)$ such that for all $x \geq x_0$ and $x^\theta \leq h \leq x$, the following hold whenever the denominators are nonzero:Let $R_{h,k}(I) := \pi_h(I)/\pi_k(I)$ and define the computable bounds:

$$L_{h,k}(I) := (\mu_h^-(I)/\mu_k^+(I)) \cdot (A_h(I)/A_k(I)), \quad U_{h,k}(I) := (\mu_h^+(I)/\mu_k^-(I)) \cdot (A_h(I)/A_k(I))$$

Then for $(h, k) \in \{(2, 4), (2, 6), (4, 6)\}$ one has the two-variable ratio sandwich:

$$\kappa^{-1} L_{h,k}(I) \leq R_{h,k}(I) \leq \kappa U_{h,k}(I).$$

Conjecture A.43 (Semiprime proxy (prime-free))**.** For $Q_r(I) := \#\{n \in I : n \text{ is semiprime and } n \equiv r \pmod{30}\}$, set $\tilde{P}_r(I) := \sqrt{Q_r(I)}$, and define $\tilde{A}_h, \tilde{\mu}_h^\pm$ analogously. There exists $\kappa_Q \geq 1$ such that:

$$\kappa_Q^{-1} \tilde{L}_{h,k}(I) \leq R_{h,k}(I) \leq \kappa_Q \tilde{U}_{h,k}(I)$$

where $\tilde{L}_{h,k}, \tilde{U}_{h,k}$ are defined from $\tilde{P}_r$ exactly as $L_{h,k}, U_{h,k}$ are from P_r.

Setup. For $k \geq 2$, let $\Omega(n)$ be the total number of prime factors of n with multiplicity. For an interval $I = (x, x + h] \subset \mathbb{R}$, define the k-almost-prime indicator $A_k(n) := 1_{\Omega(n)=k}$. For an even gap g, define the pair count:

$$\Pi_{k,g}(I) := \#\{n \in \mathbb{Z} \cap I : \Omega(n) = k \text{ and } \Omega(n + g) = k\}.$$

Fix a modulus $m \geq 2$. For a residue $r \in \mathbb{Z}/m\mathbb{Z}$, set the mod-$m$ k-almost-prime marginals:

$$A_r^{(k,m)}(I) := \#\{n \in \mathbb{Z} \cap I : \Omega(n) = k \text{ and } n \equiv r \pmod{m}\}$$

For each even gap g, let the active starts be $S_g(I) := \{r \in \mathbb{Z}/m\mathbb{Z} : A_r^{(k,m)}(I) > 0 \text{ and } A_{r+g}^{(k,m)}(I) > 0\}$ and define:

$$A_g^{(k,m)}(I) := \sum_{r \in S_g(I)} A_r^{(k,m)}(I)$$

$$\mu_g^-(I) := \min_{r \in S_g(I)} A_{r+g}^{(k,m)}(I), \quad \mu_g^+(I) := \max_{r \in S_g(I)} A_{r+g}^{(k,m)}(I)$$

Bilinear proxy and elementary boundDefine the residue-bilinear proxy:

$$T_g^{(k,m)}(I) := \sum_{r \in S_g(I)} A_r^{(k,m)}(I) \cdot A_{r+g}^{(k,m)}(I)$$

Then by min–max, $\mu_g^-(I) \cdot A_g^{(k,m)}(I) \leq T_g^{(k,m)}(I) \leq \mu_g^+(I) \cdot A_g^{(k,m)}(I)$.

Conjecture A.44 (Residue-shift ratio sandwich for k-almost primes)**.** There exists, for each $k \geq 2$ and modulus m, a universal distortion $\kappa_k(m) \geq 1$ such that for any interval $I = (x, x + h]$ with $\Pi_{k,g'}(I) > 0$, and any even gaps g, g' (e.g., 2, 4, 6), the two-variable ratio:

$$R_{k;g,g'}(I) := \Pi_{k,g}(I)/\Pi_{k,g'}(I)$$

satisfies:

$$\kappa_k(m)^{-1} \cdot L_{k;g,g'}^{(m)}(I) \leq R_{k;g,g'}(I) \leq \kappa_k(m) \cdot U_{k;g,g'}^{(m)}(I)$$

where the computable envelopes are:

$$L^{(m)}_{k;g,g'}(I) := (\mu_g^-(I)/\mu_{g'}^+(I)) \cdot (A_g^{(k,m)}(I)/A_{g'}^{(k,m)}(I))$$

$$U^{(m)}_{k;g,g'}(I) := (\mu_g^+(I)/\mu_{g'}^-(I)) \cdot (A_g^{(k,m)}(I)/A_{g'}^{(k,m)}(I)).$$

Heuristic: $\Pi_{k,g}(I)$ is comparable to $T_g^{(k,m)}(I)$ (up to $\kappa_k(m)$) because the event $\{\Omega(n) = k, \Omega(n+g) = k\}$ factorizes weakly across residue fibers modulo m. The same leading singular weights appear inside each g when restricted to a fixed residue, while their cross-g variation is absorbed into the universal $\kappa_k(m)$.

A.1.6. *Matrix.* Total number of inequalities: 44.
Total incorrect inequalities: 3.
Rate of correctness: 93.18%.

Conversations	Total Conjectures	Incorrect Conjectures
Conversation 1	10	1
Conversation 2	16	0
Conversation 3	8	2
Conversation 4	2	0
Conversation 5	8	0

A.2. **On the distribution of prime k-tuples.** In this section, we present conjectures on the distribution of prime k-tuples that were not produced from simple prompts, nor from directing the agent to generate inequalities alone, unlike the conversations in the previous section. Although we provided more specific prompts, they remained brief and relatively straightforward rather than excessively detailed. We chose this experiment to observe the machine's capacity of discovering new results when given more specification. In some instances during a conversation, we asked the machine to generate any conjecture in that direction it found promising. For such cases, it has combined two different domains of mathematics and came up with quite interesting results. All raw output, including the original input prompts, are available on GitHub.

A.2.1. *Conversation 1.*

Definition A.17. Fix q with $6 \mid q$. Let $\Delta \subseteq \mathbb{Z}$ be a finite admissible tuple with $0 \in \Delta$ and $k := |\Delta| \geq 2$. Admissible means that for every prime p, the set $\{d \pmod p : d \in \Delta\}$ omits at least one residue class. Let $M(\Delta) := \max \Delta$.

- For a residue class $a \pmod q$, write $\pi(x; q, a) := \#\{p \leq x : p \text{ prime}, p \equiv a \pmod q\}$.
- For a tuple Δ and class a, define $\pi_\Delta(x; q, a) := \#\{p \leq x - M(\Delta) : p \equiv a \pmod q$ and $p + d$ is prime for all $d \in \Delta\}$.
- Define the two mod-6 fibers inside $(\mathbb{Z}/q\mathbb{Z})^\times$: $F_1(q) := \{a \pmod q : a \equiv 1 \pmod 6, \gcd(a, q) = 1\}$, and $F_5(q)$ analogously with 5. For $r \in \{1, 5\}$, define the fiber-admissible subset $F_r(\Delta; q) := \{a \in F_r(q) : \gcd(a + d, q) = 1 \text{ for all } d \in \Delta\}$.
- Aggregate fiber counts and start supplies by $\Pi_\Delta(x; q, r) := \sum_{a \in F_r(\Delta;q)} \pi_\Delta(x; q, a)$, and $S_r^\Delta(x; q) := \sum_{a \in F_r(\Delta;q)} \pi(x - M(\Delta); q, a)$.
- Let $\mathrm{Li}_k(x) := \int_2^x \frac{dt}{(\log t)^k} \sim \frac{x}{(\log x)^k}$. Let $\mathfrak{S}(\Delta)$ be the Hardy–Littlewood singular series for Δ:

$$\mathfrak{S}(\Delta) = \prod_p \left(1 - \frac{1}{p}\right)^{-k} \left(1 - \frac{\nu_p(\Delta)}{p}\right)$$

  where $\nu_p(\Delta)$ is the number of distinct residues in Δ modulo p.

Conjecture A.45. For each fixed q with $6 \mid q$ and each $r \in \{1, 5\}$, as $x \to \infty$:

(1) $\Pi_\Delta(x; q, r) = \left(\frac{|F_r(\Delta;q)|}{\phi(q)}\right) \cdot \mathfrak{S}(\Delta) \cdot \mathrm{Li}_k(x) + O_{q,\Delta}\left(\frac{x}{(\log x)^{k+1}}\right)$.

(2) $\frac{\Pi_\Delta(x;q,r)}{S_r^\Delta(x;q)} = \frac{\mathfrak{S}(\Delta)}{(\log x)^{k-1}} + O_{q,\Delta}\left(\frac{1}{(\log x)^k}\right)$.

(3) $\left|\frac{\Pi_\Delta(x;q,1)}{S_1^\Delta(x;q)} - \frac{\Pi_\Delta(x;q,5)}{S_5^\Delta(x;q)}\right| \ll_{q,\Delta} \frac{1}{(\log x)^k}$.

Setup. Fix q with $6 \mid q$. Let $\Delta \subseteq \mathbb{Z}$ be a finite admissible tuple with $0 \in \Delta$ and $k = |\Delta| \geq 2$. Write $M(\Delta) = \max \Delta$.

- Define the start-normalized observable:

$$R_\Delta(x; q, r) := (\log x)^{k-1} \cdot \frac{\Pi_\Delta(x; q, r)}{S_r^\Delta(x; q)}.$$

- The centered-and-scaled error for joint distribution is:

$$Z_\Delta(x; q, r) := (\log x) \cdot (R_\Delta(x; q, r) - \mathfrak{S}(\Delta)).$$

- For logarithmic averaging, define:

$$\langle F \rangle_X := \frac{1}{\log X} \int_2^X F(t) \frac{dt}{t}.$$

Conjecture A.46. For each fixed $q, \Delta, r \in \{1, 5\}$, $\langle |R_\Delta(\cdot; q, r) - \mathfrak{S}(\Delta)| \rangle_X \to 0$ as $X \to \infty$. Quantitatively:

$$\langle |R_\Delta(\cdot; q, r) - \mathfrak{S}(\Delta)|^m \rangle_X \ll_{q,\Delta,m} \frac{1}{(\log X)^m}.$$

Conjecture A.47 (Multivariate Logarithmic Limiting Distribution)**.** Form the vector $Z_I(x) := (Z_{\Delta_i}(x; q, r_i))_{i \in I}$. There exists a nondegenerate Borel probability measure ν_I on $\mathbb{R}^{|I|}$ such that $\langle 1_{Z_I(\cdot) \in B} \rangle_X \to \nu_I(B)$. Heuristically, ν_I is Gaussian with mean 0. In particular, each coordinate Z_Δ changes sign infinitely often with positive logarithmic lower density.

Conjecture A.48 (Gaussian moments asymptotics)**.** For any even $m \geq 2$:

$$\langle Z_{\Delta_i}(\cdot; q, r_i)^m \rangle_X \to (m-1)!! \cdot \Sigma_{ii}^{m/2}$$

Jointly, mixed moments obey Wick's rule asymptotically.

Conjecture A.49 (Explicit Covariance via Dirichlet Characters)**.** The limiting covariance takes the form:

$$\Sigma_{(\Delta,r),(\Delta',r')} = \mathrm{Re} \sum_{\chi \pmod q} W_{\Delta,r}(\chi) \cdot \overline{W_{\Delta',r'}(\chi)} \cdot \sum_{\rho_\chi} \frac{1}{1/4 + \gamma_\chi^2} \tag{A.1}$$

where $W_{\Delta,r}(\chi)$ encodes fiber signature and tuple-local sieve content.

Conjecture A.50. The inter-fiber difference $D_\Delta(x; q) := R_\Delta(x; q, 1) - R_\Delta(x; q, 5)$ changes sign infinitely often. Its positive and negative logarithmic densities exist and lie strictly between 0 and 1.

Setup and Objects.

- Fix $q \in \mathbb{N}$ with $6 \mid q$. Let $F_5(q) := \{a \pmod q : a \equiv 5 \pmod 6, \gcd(a, q) = 1\}$ and $F_1(q)$ analogously with 1.
- For twins $\Delta_2 = \{0, 2\}$ and cousins $\Delta_4 = \{0, 4\}$, define the admissible sub-fibers $F_5(\Delta_2; q)$ and $F_1(\Delta_4; q)$.

- Define the twin/cousin start-normalized local ratios: $r_2(a;x) := \frac{\pi_{\{0,2\}}(x;q,a)}{S_5(a;x)}$, $r_4(b;x) := \frac{\pi_{\{0,4\}}(x;q,b)}{S_1(b;x)}$.
- Let $H := \ell^2(F_5(\Delta_2;q)) \oplus \ell^2(F_1(\Delta_4;q))$. Define the off-diagonal operators $A_x : \ell^2(F_5) \to \ell^2(F_1)$ and $B_x : \ell^2(F_1) \to \ell^2(F_5)$ by the congruence-shift rules: $(A_x e_a) = r_2(a;x)e_{a+2}$, $\quad (B_x e_b) = r_4(b;x)e_{b+4}$.
- Assemble the block operator $J_x := \begin{pmatrix} 0 & A_x \\ B_x & 0 \end{pmatrix}$ acting on H. Let P_2, P_4 be the pure congruence permutations.

Conjecture A.51 (Asymptotic Permutation Law for Pairs)**.** As $x \to \infty$, in operator norm on H:

$$||(\log x)A_x - \kappa_2 P_2|| \to 0, \quad ||(\log x)B_x - \kappa_2 P_4|| \to 0$$

where $\kappa_2 = 2C_2$. Deviations are controlled as $\ll_q 1$.

Conjecture A.52 (Sexy-by-Composition and Anti-commutator Centrality)**.** (1) Composition to sexy: $\|(\log x)^2(B_x A_x) - \kappa_{026} P_6^{(5)}\| \to 0$ on $\ell^2(F_5)$.

(2) Anti-commutator centrality: The block anti-commutator $\{A_x, B_x\} := A_x B_x + B_x A_x$ is asymptotically central relative to the permutation algebra generated by P_2, P_4.

Conjecture A.53 (Quasi-factorization Law for Triplets)**.** For each $a \in F_5(\Delta_2;q)$ such that $a+2 \in F_1(\Delta_4;q)$, the triplet ratio $r_{026}(a;x) := \frac{\pi_{\{0,2,6\}}(x;q,a)}{\pi(x-6;q,a)}$ satisfies:

$$\frac{(\log x)^2 r_{026}(a;x)}{[(\log x)r_2(a;x) \cdot (\log x)r_4(a+2;x)]} \to \frac{\kappa_{026}}{\kappa_2^2}.$$

Conjecture A.54 (Spectral Clustering)**.** The spectrum of $K_x := (\log x)J_x$ converges (in Hausdorff distance) to the spectrum of the limit operator K. The eigenvalues of K lie on finitely many symmetric pairs $\pm\kappa_2\zeta$ determined by the cycle structure of the $+6$ shift on the residue classes.

A.2.2. *Conversation 2.* **Notation for a family of admissible tuples.**

- Let $H_1, \ldots, H_r \subset \mathbb{Z}$ be admissible k-tuples with $0 \in H_i$ and $k_i := |H_i| \geq 2$.
- $\Pi_H(t) := \#\{n \leq t : n + h \text{ is prime for all } h \in H\}$.
- For a squarefree $M \geq 2$ and $a \in U_M := (\mathbb{Z}/M\mathbb{Z})^\times$, define the feasibility mask: $J_{H,M}(a) := 1$ if (for every prime $q|M$ and every $h \in H$) $a + h \not\equiv 0 \pmod q$, and 0 otherwise.
- Short-interval prime counts in residue classes: $\Delta_{\text{AP}}(x,y;M,a) := \pi(x+y;M,a) - \pi(x;M,a)$.
- Short-interval tuple starts: $\Delta_H(x,y) := \Pi_H(x+y) - \Pi_H(x)$.
- The singular series $S(H) := \prod_p (1-1/p)^{-k}(1-\nu_p(H)/p)$, where $\nu_p(H)$ is the number of distinct residues $-h \pmod p$ hit by H. Define $\lambda_H(x) := S(H)/(\log x)^{k-1}$ (natural log).
- Stack vectors: $\Delta_{\text{vec}}(x,y) \in \mathbb{R}^r$ with i-th entry $\Delta_{H_i}(x,y)$; $\Delta_{AP}(x,y;M) \in \mathbb{R}^{|U_M|}$ with a-th entry $\Delta_{AP}(x,y;M,a)$.
- Design matrix $B(x) \in \mathbb{R}^{r \times |U_M|}$ with entries $B_{i,a}(x) := \lambda_{H_i}(x) \cdot J_{H_i,M}(a)$.
- Let $k_{\min} := \min_i k_i$.

Conjecture A.55. Fix squarefree M for which the rows $a \mapsto J_{H_i,M}(a)$ are not all identical. There exist absolute constants $x_0 \geq 10$, $\theta \in (0,1)$, $K > 0$ such that for all $x \geq x_0$ and $x^\theta \leq y \leq x$,

$$\|\Delta_{\text{vec}}(x,y) - B(x) \cdot \Delta_{\text{AP}}(x,y;M)\|_2 \leq K \cdot \sqrt{r} \cdot \sqrt{y}/(\log x)^{k_{min}}$$

Equivalently, for every weight vector $w \in \mathbb{R}^r$,

$$|w \cdot \Delta_{\text{vec}}(x,y) - \sum_{a \in U_M} (\sum_{i=1}^{r} w_i \lambda_{H_i}(x) J_{H_i,M}(a)) \cdot \Delta_{AP}(x,y;M,a)| \leq K \cdot \|w\|_2 \cdot \sqrt{y}/(\log x)^{k_{min}}.$$

A.2.3. *Conversation 3.*

Definition A.18. Let $E/\mathbb{Q}$ be an elliptic curve with good reduction outside a finite set N_E. For $p \nmid N_E$, write the trace of Frobenius as:

$$a_p(E) = p + 1 - \#E(\mathbb{Z}/p\mathbb{Z}) = 2\sqrt{p}\cos\theta_p(E) \text{ with } \theta_p(E) \in [0, \pi]$$

Fix modulus $m = 30$ and the admissible odd residues $R = \{1, 7, 11, 13, 17, 19, 23, 29\}$. For an interval $I = (x, x+h]$ with $x^\theta \le h \le x$ (fixed $\theta \in (0,1)$), and a finite partition of $[0, \pi]$ into measurable bins $\mathfrak{B} = \{B_1, \ldots, B_s\}$, define:

- $P_{r,i}(I; E) := \#\{\text{primes } p \in I : p \equiv r \pmod{30}, \theta_p(E) \in B_i\}$.
- For even gaps $g \in \{2, 4, 6\}$, let $S_g = \{11, 17, 29\}$ if $g = 2$; $\{7, 13, 19\}$ if $g = 4$; $\{1, 7, 11, 13, 17, 23\}$ if $g = 6$.Angle-conditioned pair counts: $\pi_g^E(I; i, j) := \#\{\text{primes } p \in I : p, p{+}g \text{ are prime}, \theta_p(E) \in B_i, \theta_{p+g}(E) \in B_j\}$.
- Angle–residue bilinear proxy: $T_{g;i,j}(I; E) := \sum_{r \in S_g} P_{r,i}(I; E) \cdot P_{r+g,j}(I; E)$.
- Aggregates and extrema: $A_{g,i}(I; E) := \sum_{r \in S_g} P_{r,i}(I; E)$, $\mu_{g,j}^-(I; E) := \min_{r \in S_g} P_{r+g,j}(I; E)$, $\mu_{g,j}^+(I; E) := \max_{r \in S_g} P_{r+g,j}(I; E)$.

Elementary min–max gives $\mu_{g,j}^- \cdot A_{g,i} \le T_{g;i,j} \le \mu_{g,j}^+ \cdot A_{g,i}$.

Conjecture A.56. There exists $\kappa_E \ge 1$, depending only on E (and not on x, h or the bins), such that for any $(g, k) \in \{(2,4), (2,6), (4,6)\}$ and any i, j with $\pi_k^E(I; i, j) > 0$:

$$\kappa_E^{-1} \cdot L_{g,k;i,j}(I; E) \le \frac{\pi_g^E(I; i, j)}{\pi_k^E(I; i, j)} \le \kappa_E \cdot U_{g,k;i,j}(I; E)$$

where the observable envelopes are:

$$L_{g,k;i,j} := \left(\frac{\mu_{g,j}^-}{\mu_{k,j}^+}\right) \cdot \left(\frac{A_{g,i}}{A_{k,i}}\right), \quad U_{g,k;i,j} := \left(\frac{\mu_{g,j}^+}{\mu_{k,j}^-}\right) \cdot \left(\frac{A_{g,i}}{A_{k,i}}\right).$$

Conjecture A.57. Let $\mu_{\mathrm{ST}}(B) := \frac{2}{\pi} \int_B \sin^2\theta \, d\theta$ be the Sato–Tate mass (for non-CM E). Then there exists $\kappa'_E \ge 1$ such that for each fixed $g \in \{2, 4, 6\}$ and any bins i, j with $\pi_g(I) > 0$:

$$\kappa'^{-1}_E \le \frac{\pi_g^E(I; i, j)}{\mu_{\mathrm{ST}}(B_i)\mu_{\mathrm{ST}}(B_j) \cdot \pi_g(I)} \le \kappa'_E.$$

A.2.4. *Conversation 4.*

Conjecture A.58. Let $x \to \infty$ and let $a_n \ge 0$ be weights supported on $I_x := (x/2, x]$ with average $1 + o(1)$:

$$A(x) := \frac{2}{x} \sum_{x/2 < n \le x} a_n = 1 + o(1)$$

Let b_n be a comparably bounded "structured approximation" to a_n (e.g., a well-factorable or Dirichlet-convolution model at some level), and set $w_n := a_n - b_n$. Fix parameters $0 < \gamma < 1$, $0 \le \theta \le 1/2$, and $0 < \nu < 1 - \theta$. Assume divisor-boundedness: $a_n, b_n \ll_\epsilon x^\epsilon$, and that the following hold uniformly in x.

(1) Type I (strong mean-zero in progressions to level x^θ with L^2-saving x^γ):

$$\sum_{q \le x^\theta} \sum_{(a,q)=1} \left| \sum_{n \in I_x, n \equiv a \pmod{q}} w_n \right|^2 \ll x^{2-2\gamma}$$

(Equivalently, on average over $q \le x^\theta$, the progression sums of w_n exhibit power cancellation with exponent γ.)

(2) Type II (bilinear cancellation in the balanced ranges):For all M, N with $x^{\nu} \leq M, N \leq x^{1-\nu}$ and $MN \sim x$, and all complex sequences α_m, β_n supported on $m \sim M, n \sim N$ with $\|\alpha\|_2 = \|\beta\|_2 = 1$ and $|\alpha_m|, |\beta_n| \ll_{\epsilon} (mn)^{\epsilon}$:

$$B(M, N; \alpha, \beta) := \left| \sum_{m \sim M} \sum_{n \sim N} w_{mn} \alpha_m \beta_n \right| \ll x^{1-\gamma}$$

(Thus the bilinear form in w_{mn} has power-saving x^{γ} uniformly across the bilinear rectangles.)

Conjecture A.59 (Phase diagram for primes)**.** Assume Type I and Type II as above. If

$$\theta + 2\nu > 1$$

then one has the prime asymptotic with a power-saving error governed by γ:

$$\sum_{x/2 < p \leq x} a_p = (1 + o(1)) \frac{x}{\log x}$$

more quantitatively:

$$\sum_{x/2 < p \leq x} a_p = \frac{x}{\log x} + O\left(\frac{x^{1 - c \cdot \min(\gamma, \theta + 2\nu - 1)}}{\log x} \right)$$

for some absolute $c > 0$ (independent of x), provided b_n is calibrated so that $\sum_{x/2 < n \leq x} b_n \Lambda(n)$ reproduces the expected main term up to $o(x/\log x)$.

Setup. Let $x \to \infty$. Let nonnegative weights a_n be supported on $I_x := (x/2, x]$, satisfy divisor-boundedness $a_n \ll_{\epsilon} x^{\epsilon}$, and have normalized average $A(x) := (2/x) \sum_{x/2 < n \leq x} a_n = 1 + o(1)$. Let b_n be a structured model with $b_n \ll_{\epsilon} x^{\epsilon}$ such that $\sum_{x/2 < n \leq x} b_n \Lambda(n) = x/\log x + o(x/\log x)$. Put $w_n := a_n - b_n$.

Conjecture A.60. Fix $0 < \gamma_I, \gamma_{II} < 1$, $0 \leq \theta \leq 1/2$, $0 < \nu < 1 - \theta$, and $0 \leq \kappa < 1$.

- Type I (L^2 cancellation in progressions up to x^{θ} with saving x^{γ_I}):

$$S_I(x) := \sum_{q \leq x^{\theta}} \sum_{(a,q)=1} \left| \sum_{n \in I_x, n \equiv a \pmod{q}} w_n \right|^2 \ll x^{2 - 2\gamma_I}.$$

- Type II (bilinear cancellation with sparse exceptional rectangles): For every dyadic M, N with $MN \sim x$ and $\min(M, N) \geq x^{\nu}$, for all sequences α_m, β_n supported on $m \sim M, n \sim N$ with $\|\alpha\|_2 = \|\beta\|_2 = 1$ and $|\alpha_m|, |\beta_n| \ll_{\epsilon} (mn)^{\epsilon}$, one has

$$B(M, N; \alpha, \beta) := \left| \sum_{m \sim M} \sum_{n \sim N} w_{mn} \alpha_m \beta_n \right| \leq x^{1 - \gamma_{II}}$$

except on a set E_x of dyadic rectangles whose logarithmic-area density is $\leq x^{-\kappa}$:

$$\mathrm{dens}_{\log}(E_x) := \frac{|\{(\log M, \log N) : MN \sim x, \min(M, N) \geq x^{\nu}, \text{ and } (M, N) \text{ exceptional}\}|}{|\{\text{all such } (M, N)\}|} \leq x^{-\kappa}.$$

Conclusion (Robust phase diagram). If

$$\theta + 2\nu > 1 + \kappa$$

then there exists $c = c(\epsilon) > 0$ such that

$$\sum_{x/2 < p \leq x} a_p = \frac{x}{\log x} + O\left(\frac{x^{1 - c \cdot \min(\gamma_I, \gamma_{II}, \theta + 2\nu - 1 - \kappa)}}{\log x} \right)$$

Borderline. At $\theta + 2\nu = 1 + \kappa$, the same holds with a power-saving replaced by a logarithmic loss in the error (polylog factors), assuming mild well-factorability of b_n.

Setup. Fix $0 < \rho < 1/2$ and an irrational $\alpha \in \mathbb{R}, \beta \in \mathbb{R}/\mathbb{Z}$. For $X \to \infty$ define the Bohr-window prime counting function on the dyadic interval $(X/2, X]$:

$$S_{\alpha,\beta,\rho}(X) := \#\{X/2 < p \leq X : p \text{ prime}, \|\alpha p - \beta\| < p^{-\rho}\}.$$

Model weights and cancellation. Let a_n be nonnegative weights supported on $(X/2, X]$ that majorize the indicator of the Bohr window and satisfy divisor-boundedness $a_n \ll_\epsilon X^\epsilon$ with normalized average $1+o(1)$. Let b_n be a structured approximation (well-factorable/sieved proxy) with $b_n \ll_\epsilon X^\epsilon$ reproducing the main term for Λ via $\sum_{X/2<n\leq X} b_n\Lambda(n) = X/\log X + o(X/\log X)$. Put $w_n := a_n - b_n$. Fix exponents $0 < \gamma < 1, 0 \leq \theta \leq 1/2, 0 < \nu < 1 - \theta$. Assume the following uniform bounds:

- Type I (L^2 mean-zero in APs to level X^θ with saving X^γ):

$$\sum_{q\leq X^\theta} \sum_{(a,q)=1} \left| \sum_{X/2<n\leq X, n\equiv a \pmod q} w_n \right|^2 \ll X^{2-2\gamma}.$$

- Type II (bilinear cancellation across $MN \approx X$ with min-length $\geq X^\nu$ and saving X^γ): For all dyadic M, N with $MN \approx X$ and $\min(M, N) \geq X^\nu$, and all α_m, β_n with $\|\alpha\|_2 = \|\beta\|_2 = 1$ and $|\alpha_m|, |\beta_n| \ll_\epsilon (mn)^\epsilon$:

$$B(M, N; \alpha, \beta) := \left| \sum_{m\sim M} \sum_{n\sim N} w_{mn}\alpha_m\beta_n \right| \ll X^{1-\gamma}.$$

- Bohr truncation. Approximate $1_{\|\alpha n-\beta\|<n^{-\rho}}$ by a Vaaler trigonometric polynomial of height $H \approx X^\rho$:

$$1_{\|t\|<\delta} = \sum_{|h|\leq H} c_h(\delta, H)e(ht) + O(H^{-1}), \text{ with } \sum_{|h|\leq H} |c_h| \ll 1$$

  transported to $t = \alpha n - \beta$ and $\delta = n^{-\rho}$, absorbing the $O(H^{-1})$ error into b_n.

Conjecture A.61 (Bohr-window phase law)**.** If

$$\theta + 2\nu > 1 - 2\rho$$

then there exists $c > 0$ such that

$$S_{\alpha,\beta,\rho}(X) = (2 + o(1)) \sum_{X/2<p\leq X} p^{-\rho} + O\left(\frac{X^{1-c\cdot\min(\gamma,\theta+2\nu-(1-2\rho))}}{\log X}\right)$$

Equivalently, by partial summation:

$$S_{\alpha,\beta,\rho}(X) = \left(\frac{2}{1-\rho} + o(1)\right)\frac{X^{1-\rho}}{\log X} + O\left(\frac{X^{1-c\cdot\min(\gamma,\theta+2\nu-(1-2\rho))}}{\log X}\right).$$

Borderline. At $\theta + 2\nu = 1 - 2\rho$, one expects only logarithmic savings in the error (polylog losses) unless additional well-factorability of b_n is imposed.

Setup. Fix an integer base $b \geq 2$ and a digit $\delta \in \{0, 1, \ldots, b-1\}$. Let $S_{b,\delta}$ be the set of $n \geq 1$ whose base-b expansion avoids the digit δ. For $X \to \infty$, consider the prime counting function in a dyadic interval:

$$\Pi_{b,\delta}(X) := \#\{X/2 < p \leq X : p \text{ is prime}, n := p \in S_{b,\delta}\}$$

The set $S_{b,\delta}$ has size $\asymp X^{\rho_b}$ with $\rho_b := \log_b(b-1)$. We expect the main term size $X^{\rho_b}/\log X$. *Weights and Structured Model.* Let a_n be a nonnegative digital weight majorizing/minorizing $1_{S_{b,\delta}}$ on $(X/2, X]$, with divisor-boundedness $a_n \ll_\epsilon X^\epsilon$, and whose Walsh–Fourier expansion is truncated at the natural spectral height $H \asymp X^{\sigma_b}$ where $\sigma_b := 1 - \rho_b = \log_b(b/(b-1))$. Let b_n be a well-factorable/sieved structured approximation such that:

$$\sum_{X/2<n\le X} b_n\Lambda(n) = C_{b,\delta}\frac{X^{\rho_b}}{\log X} + o\left(\frac{X^{\rho_b}}{\log X}\right)$$

Set $w_n := a_n - b_n$.

Conjecture A.62. Assume the following hold uniformly over all Walsh frequencies $|h| \lesssim H$:Type I (L^2 Mean-Zero in Progressions):

$$\sum_{q\le X^\theta}\sum_{(a,q)=1}\left|\sum_{n\in I_X, n\equiv a \pmod q} w_n\cdot\omega_h(n)\right|^2 \ll X^{2-2\gamma}$$

Type II (Bilinear Cancellation): For $MN \asymp X$ and $\min(M,N) \ge X^\nu$:

$$B(M,N;\alpha,\beta,h) := \left|\sum_{m\sim M}\sum_{n\sim N} w_{mn}\cdot\omega_h(mn)\alpha_m\beta_n\right| \ll X^{1-\gamma}.$$

Conjecture A.63. If $\gamma+\theta+2\nu>1$, then:

$$\Pi_{b,\delta}(X) = C_{b,\delta}\frac{X^{\rho_b}}{\log X} + O\left(\frac{X^{\rho_b - c\cdot\min(\gamma,\gamma+\theta+2\nu-1)}}{\log X}\right).$$

Setup. Let $X\to\infty$. Consider the thin set $A_X = \{n \in (X/2, X] : n = x^2+(y^2+1)^2 \text{ for some } x, y \in \mathbb{Z}\}$.Let $a_n \ge 0$ be divisor-bounded weights ($a_n \ll_\epsilon X^\epsilon$) supported on $(X/2, X]$ that majorize/minorize 1_{A_X} with smooth cutoffs in x, y at scales $X^{1/2}$ and $X^{1/4}$ respectively, with normalized average $1+o(1)$.Let b_n be a structured approximation (well-factorable/sieved model) with $b_n \ll_\epsilon X^\epsilon$ calibrated so that:

$$\sum_{X/2<n\le X} b_n\Lambda(n) = C\cdot\frac{X^{3/4}}{\log X} + o\left(\frac{X^{3/4}}{\log X}\right)$$

for some singular series constant $C>0$. Put $w_n := a_n - b_n$. *Uniform Quartic-Frequency Control.* Writing the indicator of the inner constraint through a δ-method or van der Corput–Weyl expansion introduces quartic additive twists $e(h(y^2+1)^2)$ with frequency height $H \asymp X^{1/4}$. Assume uniformity of the following bounds for all $|h| \le H$:

- Type I (L^2 cancellation in APs to level X^θ with saving X^γ):

  $$S_I(X;h) := \sum_{q\le X^\theta}\sum_{(a,q)=1}\left|\sum_{n\in I_x, n\equiv a \pmod q} w_n\cdot\omega_h(n)\right|^2 \ll X^{2-2\gamma}$$

  where ω_h encodes the quartic additive twist transported to n.
- Type II (Bilinear cancellation for $MN \asymp X$ with min-length $\ge X^\nu$ and saving X^γ):For all dyadic M, N with $MN \asymp X$ and $\min(M,N) \ge X^\nu$, and all coefficient sequences α_m, β_n with $\|\alpha\|_2 = \|\beta\|_2 = 1$ and $|\alpha_m|, |\beta_n| \ll_\epsilon (mn)^\epsilon$:

  $$B(M,N;\alpha,\beta,h) := \left|\sum_{m\sim M}\sum_{n\sim N} w_{mn}\cdot\omega_h(mn)\alpha_m\beta_n\right| \ll X^{1-\gamma}$$

  uniformly for $|h| \le H$.

Conjecture A.64 (γ-shifted Phase Inequality)**.** If $\gamma + \theta + 2\nu > 1$, then there exists $c = c(\epsilon) > 0$ such that the prime count in $(X/2, X]$ satisfies:

$$\Pi(X) := \#\{X/2 < p \le X : p = x^2 + (y^2+1)^2\} = C \cdot \frac{X^{3/4}}{\log X} + O\left(\frac{X^{3/4 - c\cdot\min(\gamma, \gamma+\theta+2\nu-1)}}{\log X}\right)$$

Setup. Let $x \to \infty$. Consider the thin set $A_x := \{n \in (x/2, x] : n = x^2 + (y^3+z^3)^2 \text{ for some integers } x, y, z\}$.

- Weights: Let $a_n \ge 0$ be divisor-bounded weights ($a_n \ll_\epsilon x^\epsilon$) supported on $(x/2, x]$ that majorize/minorize 1_{A_x} with smooth cutoffs $x \asymp x^{1/2}$ and $y, z \asymp x^{1/6}$.
- Average: Normalized average $A(x) := (2/x) \sum_{x/2<n\le x} a_n = 1 + o(1)$.
- Model: Let b_n be a structured approximation (well-factorable/sieved proxy) calibrated so that $\sum_{x/2<n\le x} b_n \Lambda(n) = C \cdot x^{5/6}/\log x + o(x^{5/6}/\log x)$.
- Error Weight: Set $w_n := a_n - b_n$.
- Nested Harmonic Structure: Implement the constraints via a two-level δ-method: $n = x^2 + t^2$ with $t \asymp x^{1/2}$ and $t = y^3 + z^3$ with $y, z \asymp x^{1/6}$. The inner cubic layer introduces additive twists $\omega_h(n)$ from phases $e(h \cdot (y^3 + z^3))$, with effective harmonic height $H \asymp x^{1/6+o(1)}$. Assume uniformity for all $|h| \le H$:
  1. Type I (L^2 cancellation in APs): $\sum_{q\le x^\theta} \sum_{(a,q)=1} |\sum_{x/2<n\le x, n\equiv a \pmod q} w_n \cdot \omega_h(n)|^2 \ll x^{2-2\gamma}$
  2. Type II (Bilinear cancellation):For dyadic M, N with $MN \approx x$ and $\min(M, N) \ge x^\nu$:$B(M, N; \alpha, \beta, h) := |\sum_{m\sim M} \sum_{n\sim N} w_{mn} \cdot \omega_h(mn) \alpha_m \beta_n| \ll x^{1-\gamma}$

Conjecture A.65 (γ-shifted Phase Law)**.** If $\gamma + \theta + 2\nu > 1$, then there exists $c = c(\epsilon) > 0$ such that:

$$\Pi(x) := \#\{x/2 < p \le x : p = x^2 + (y^3+z^3)^2\} = C \frac{x^{5/6}}{\log x} + O\left(\frac{x^{5/6 - c\cdot\min(\gamma, \gamma+\theta+2\nu-1)}}{\log x}\right).$$

Setup. Fix a nonzero $a \in \mathbb{Z}$ that is not a square. For a large parameter X, let:

$$S_a(X; I) := \#\{X/2 < p \le X : p \text{ prime}, \exists x \in \mathbb{Z} \text{ with } 0 \le x < p, x^2 \equiv a \pmod p, \text{ and } x/p \in I_p\}$$

where for each prime p we are given an interval $I_p \subset [0, 1)$ of length $|I_p| \asymp p^{-2/3}$ (equivalently, the absolute length in $[0, p)$ is $\asymp p^{1/3}$). *Model Weights and Fourier Truncation in* $x \pmod p$*.* Let a_n be nonnegative, divisor-bounded weights $a_n \ll_\epsilon X^\epsilon$ supported on $(X/2, X]$ that majorize/minorize the indicator of the event "$\exists x \pmod n$ with $x^2 \equiv a \pmod n$ and $x/n \in I_n$" via a Vaaler/Walsh-type expansion of $1_{x/n\in I_n}$ on $\mathbb{Z}/n\mathbb{Z}$. This yields a decomposition over harmonics $|h| \le H_n$ with $H_n \asymp n^{2/3}$:

$$1_{x/n\in I_n} = \sum_{|h|\le H_n} c_h(n) e(hx/n) + O(H_n^{-1}), \quad \text{with} \quad \sum_{|h|\le H_n} |c_h(n)| \ll 1$$

Transporting this into the prime-detection sum produces a twisted weight sequence on n with quadratic additive trace-function factors $\omega_h(n)$ (Salié-type phases coming from summing over the roots $x \pmod n$). *Structured Proxy and Cancellation.* Choose a structured proxy b_n with $b_n \ll_\epsilon X^\epsilon$ so that:

$$\sum_{X/2<n\le X} b_n \Lambda(n) = (C_I + o(1)) X^{1/3}/\log X$$

where C_I is the expected density constant. Put $w_n := a_n - b_n$. *Uniform Type I/II Hypotheses.* Fix exponents $0 < \gamma \le 1/2$, $0 \le \theta \le 1/2$, $0 < \nu < 1 - \theta$. Assume that for all harmonics $|h| \le X^{2/3+o(1)}$ one has uniformly in h:

- Type I: $\sum_{q\le X^\theta}\sum_{(a,q)=1}|\sum_{X/2<n\le X, n\equiv a \pmod q} w_n\cdot\omega_h(n)|^2 \ll X^{2-2\gamma}$.
- Type II: $B(M,N;\alpha,\beta;h) := |\sum_{m\sim M}\sum_{n\sim N} w_{mn}\cdot\omega_h(mn)\alpha_m\beta_n| \ll X^{1-\gamma}$.

Conjecture A.66. If $\gamma+\theta+2\nu>1$, then for some absolute $c>0$:

$$S_a(X;I) = C_I\cdot\frac{X^{1/3}}{\log X} + O\left(\frac{X^{1/3-c\cdot\min(\gamma,\gamma+\theta+2\nu-1)}}{\log X}\right).$$

A.2.5. *Conversation 5.*

Definition A.19. Define:

- Parameters: let $x\to\infty$, choose $R=x^{1/2-\varepsilon}$ with fixed $\varepsilon>0$ and a factorization $R=R_1R_2R_3$. Work with odd squarefree moduli throughout; handle $p=2$ separately via a trivial kernel.
- Triply well-factorable base sequence: define a multiplicative $\alpha:\mathbb{N}\to\mathbb{R}$ with support on odd squarefree $d\le R$ and $\alpha=\rho_1*\rho_2*\rho_3$, where each ρ_j is supported on odd squarefree $d\le R_j$, and for odd primes $p\le R_j$, $\rho_j(p)=c_jt_p$ with $t_p:=1/(p-2)$ and constants $c_j>0$ tuned so that $\sum_{d\le R}\alpha(d)^2\asymp 1$ (normalization). For $p=2$ set $\rho_j(2)=0$.
- Local coupling kernel: for each prime p define a 2×2 kernel on $a,b\in\{0,1\}$ ($a=1_{p|d}, b=1_{p|e}$): $\kappa_p(a,b):=1-(1+t_p)(a+b)+(1+2t_p)ab$ for $p\ge 3$, and $\kappa_2(a,b):=1-a-b$. Note: $\kappa_p(1,1)=0$ for $p\ge 3$; $\kappa_p(1,0)=\kappa_p(0,1)=-t_p$; $\kappa_p(0,0)=1$.
- Bivariate coefficient: for odd squarefree d,e set $K(d,e):=\prod_{p\ge 3}\kappa_p(1_{p|d},1_{p|e})$, and define $\lambda_{d,e}:=\mu(d)\mu(e)\alpha(d)\alpha(e)K(d,e)1_{(d,e)=1}1_{2\nmid de}$. Equivalently, using $1_{(d,e)=1}=\sum_{m|(d,e)}\mu(m)$, for odd squarefree d,e one has the factorized form: $\lambda_{d,e}=\sum_{m|(d,e)}\mu(m)g(m)f(d/m)f(e/m)$, where for odd squarefree u, $f(u):=\mu(u)\alpha(u)\prod_{p|u}(-t_p)$, and $g(m):=\prod_{p|m}(1+2t_p)$. This representation exposes a three-block structure (common factor m, and the coprime residues $d/m, e/m$), compatible with triple well-factorability by assigning each block to a distinct ρ_j-scale.
- New sieve weight: for $n\le x$ define $W(n):=\sum_{d|n}\sum_{e|n+2}\lambda_{d,e}$.

Heuristics. By multiplicativity of κ_p and the calibration $t_p=1/(p-2)$, the local p-factor of the first moment of $W(n)$ matches the Hardy–Littlewood twin local factor, yielding the main term proportional to $2C_2$, where $C_2=\prod_{p\ge 3}(1-1/(p-1)^2)$.

Conjecture A.67 (pairwise triply well-factorable weight)**.** There exists $\delta>0$ such that for the weight $W(n)$ above, one has the two-dimensional dispersion bound

$$\sum_{q\le x^{1/2+\delta}}\max_{(a,q)=1}\left|\sum_{n\le x; n\equiv a \pmod q} W(n) - \frac{2C_2x}{\varphi(q)(\log x)^2}\right| \ll \frac{x}{(\log x)^A}$$

for any fixed $A>0$, uniformly in x (with constants depending on A and ε). Consequently,$\sum_{n\le x}W(n)=(2C_2+o(1))x/(\log x)^2$,and, by positivity optimization within the Selberg framework (standard truncations and smoothing), this implies a positive lower bound of the expected order for the weighted twin prime count. Under a two-dimensional Bombieri–Vinogradov hypothesis at level $\theta>1/2$ (e.g., a generalization of the Maynard–Lichtman–Pascadi triply well-factorable exponent 66/107 to pairs), this yields an unweighted lower bound$\#\{n\le x : n \text{ and } n+2 \text{ prime}\}\gg x/(\log x)^2$.

Definition A.20. Define:

- A complex sequence $(\lambda_q)_{q\le Q}$ is triply well-factorable of level Q if for any $Q_1,Q_2,Q_3\ge 1$ with $Q_1Q_2Q_3=Q$, there exist 1-bounded sequences β_j supported on $q\le Q_j$ ($j=1,2,3$)

such that, for all $q \le Q$,

$$\lambda_q = \sum_{abc=q} \beta_1(a)\beta_2(b)\beta_3(c).$$

- Pairwise extension (pair Dirichlet convolution)For functions F, G on pairs of positive integers, define the pair Dirichlet convolution by

  $$(F \star G)(d,e) := \sum_{a|d} \sum_{b|e} F(a,b)G(d/a, e/b)$$

  We say $\Lambda(d,e)$ is triply well-factorable of pair-level R if for any $R_1R_2R_3 = R$ there are 1-bounded pair-sequences B_j, supported on $\operatorname{lcm}(d,e) \le R_j$, such that $\Lambda = B_1 \star B_2 \star B_3$.
- New twin-optimized pair weight
  - Parameters: let $x \to \infty$. Fix $\varepsilon > 0$ and $R = x^{1/2-\varepsilon}$. Choose a factorization $R = R_1R_2R_3$.
  - Base univariate triply well-factorable sequence: choose $\alpha = \rho_1 * \rho_2 * \rho_3$, where each ρ_j is 1-bounded, supported on odd squarefree $d \le R_j$, and $\rho_j(2) = 0$.
  - Local damping (decoupled, twin-calibrated): for $p \ge 3$ set $t_p := 1/(p-2) \in (0,1]$, and define $u(n) := \prod_{p|n, p\ge 3}(1-t_p)$, with $u(1) = 1$, so that $0 \le u(n) \le 1$ and u is completely multiplicative on squarefree n.
  - Coprimality layer: write $1_{(d,e)=1} = \sum_{m|(d,e)} \mu(m)$.
  - Bivariate coefficient: for odd squarefree d, e,

    $$\lambda_{d,e} := \mu(d)\mu(e)\alpha(d)\alpha(e)u(d)u(e)1_{(d,e)=1}1_{2\nmid de}1_{\operatorname{lcm}(d,e)\le R}.$$

    Equivalently (exposing a three-block structure): $\lambda_{d,e} = \sum_{m|(d,e)} \mu(m)F(d/m)F(e/m)$, where $F(n) := \mu(n)\alpha(n)u(n)1_{2\nmid n}1_{n\le R}$. This splits combinatorially into three short blocks: the common factor m and the disjoint tails d/m and e/m. It can be realized as a pair triple-convolution $\Lambda = B_1 \star B_2 \star B_3$ with 1-bounded building blocks by distributing the three univariate ρ_j across the two tails and the diagonal coprimality layer.
  - Sieve weight: for $n \le x$, $W(n) := \sum_{d|n} \sum_{e|n+2} \lambda_{d,e}$.

Heuristic main term and local calibration. The multiplicative damping $u(n)$ imposes a negative bias at each $p \ge 3$ against having $p|d$ or $p|e$ (single-hit penalty), matching the local combinatorics of the twin obstruction while keeping all local multipliers in $[0,1]$.

With α supported on squarefree moduli up to $R = x^{1/2-\varepsilon}$, the first moment satisfies the formal identity $E_p :=$ average over $n \pmod p$ of contributions at $p \sim 1 - 2/(p-1) + O(p^{-2})$, and hence, across primes, the first moment of $W(n)$ matches the twin singular series up to $o(1)$, yielding:

$$\sum_{n\le x} W(n) \sim (2C_2)\frac{x}{(\log x)^2}, \text{ where } C_2 = \prod_{p\ge 3}\left(1 - \frac{1}{(p-1)^2}\right).$$

Conjecture A.68 (two-dimensional dispersion for triply well-factorable pair weights)**.** There exists $\delta = \delta(\varepsilon) > 0$ such that for the weight $W(n)$ constructed above,

$$\sum_{q\le x^{1/2+\delta}} \max_{(a,q)=1} \left| \sum_{n\le x, n\equiv a \pmod q} W(n) - \frac{2C_2x}{\varphi(q)(\log x)^2} \right| \ll_A \frac{x}{(\log x)^A}$$

for all fixed $A > 0$. Consequently, $\sum_{n\le x} W(n) = (2C_2 + o(1))x/(\log x)^2$, so that, after the standard Selberg-positivity optimization over (ρ_j), one obtains a positive lower bound of order $x/(\log x)^2$ for a suitably truncated weighted twin-prime count. Under a two-dimensional Bombieri–Vinogradov hypothesis for the pair sequence at some level $\theta > 1/2$, this yields:

$$\#\{n \le x : n \text{ and } n+2 \text{ are prime}\} \gg \frac{x}{(\log x)^2}.$$

Conjecture A.69. here exists $\delta > 0$ (depending on ε) such that, uniformly for x large,

$$\sum_{q \le x^{1/2+\delta}} \max_{(a,q)=1} \left| \sum_{n \le x, n \equiv a \pmod q} W(n) - \frac{2C_2 x}{\varphi(q)(\log x)^2} \right| \ll_A \frac{x}{(\log x)^A} \text{ for every fixed } A > 0$$

Consequently, $\sum_{n \le x} W(n) = (2C_2 + o(1))x/(\log x)^2$. Optimizing the ρ_j in a Selberg-positivity framework yields a positive lower bound of order $x/(\log x)^2$ for a truncated weighted twin count. Under a two-dimensional Bombieri–Vinogradov hypothesis at level $\theta > 1/2$ for pair sequences supported by triply well-factorable moduli, this implies the unweighted bound:

$$\#\{n \le x : n \text{ and } n+2 \text{ prime}\} \gg \frac{x}{(\log x)^2}.$$

A.2.6. *Conversation 6.*

Conjecture A.70. Let $\epsilon > 0$ and $A > 0$. For large x, set $Q \le x^{2/3-\epsilon}$. Let $\lambda = (\lambda_q)_{q \ge 2}$ be a triple well-factorable sequence of level Q, i.e., there exist functions $\beta_i : \mathbb{N} \to \mathbb{C}$ $(i = 1, 2, 3)$ with:supp $\beta_i \subset (Q_i, 2Q_i]$ and $||\beta_i||_\infty \ll (\log Q)^{O(1)}$,$Q_1 Q_2 Q_3 = Q$,$\lambda = \beta_1 * \beta_2 * \beta_3$ (Dirichlet convolution over natural numbers),$\lambda_q = 0$ unless $2 \le q \le Q$.Write the discrepancy for the cousin-pair correlation in the residue class $a \pmod q$ as:

$$\Delta(x; q, a; 4) := \sum_{n \le x, n \equiv a \pmod q} \Lambda(n)\Lambda(n+4) - c(q, a; 4)x$$

where Λ is the von Mangoldt function, and $c(q, a; 4)$ is the predicted Hardy–Littlewood main-term density restricted to the class $a \pmod q$. Concretely, one may take:

$$c(q, a; 4) := \left(\frac{\mathfrak{S}(4)}{\varphi(q)}\right) \cdot \mathbf{1}_{(a,q)=1} \cdot \mathbf{1}_{(a+4,q)=1}$$

where $\mathfrak{S}(4) = 2\prod_{p>2} \frac{p(p-2)}{(p-1)^2}$ is the singular series for the shift 4; more refined local correction factors at primes dividing q can be inserted without changing the statement below.Then, for any $A > 0$:

$$\sum_{2 \le q \le Q} |\lambda_q| \cdot \max_{a \pmod q} |\Delta(x; q, a; 4)| \ll_{A,\epsilon} x(\log x)^{-A}$$

Equivalently, uniformly over all triple-well-factorable sequences of level Q with well-factorable norm $O(1)$, the weighted average discrepancy for the pair-correlation in AP up to level $x^{2/3-\epsilon}$ exhibits power-saving cancellation.

Conjecture A.71 (Moduli Density)**.** For all but $o_\epsilon(Q)$ of such triple-well-factorable moduli $q \le Q$, there exists a residue class $a \pmod q$ with $(a(a+4), q) = 1$ such that:

$$\sum_{n \le x, n \equiv a \pmod q} \Lambda(n)\Lambda(n+4) = c(q, a; 4)x + O(x(\log x)^{-A}).$$

Conjecture A.72 (Strengthened Variance Form)**.** For the same λ and Q, one has:

$$\sum_{2 \le q \le Q} |\lambda_q|^2 \sum_{a \pmod q} |\Delta(x; q, a; 4)|^2 \ll_{A,\epsilon} x^2(\log x)^{-A}.$$

Conjecture A.73. Fix $\epsilon, \varpi > 0$ and $A > 0$. Let X be large and let $x \in [X, 2X]$, and y satisfy $X^{2/3+\varpi} \leq y \leq X$. Let $\lambda^{(1)} = (\lambda_q^{(1)})_{q \geq 2}$ and $\lambda^{(2)} = (\lambda_r^{(2)})_{r \geq 2}$ be triple well-factorable sequences of levels Q_1 and Q_2, respectively: for $i = 1, 2$ there exist $\beta_j^{(i)} : \mathbb{N} \to \mathbb{C}$ $(j = 1, 2, 3)$ such that:supp $\beta_j^{(i)} \subset (Q_{i,j}, 2Q_{i,j}]$ with $\|\beta_j^{(i)}\|_\infty \ll (\log X)^{O(1)}$,$Q_{i,1}Q_{i,2}Q_{i,3} = Q_i$, and $\lambda^{(i)} = \beta_1^{(i)} * \beta_2^{(i)} * \beta_3^{(i)}$ (Dirichlet convolution),$\lambda_m^{(i)} = 0$ unless $2 \leq m \leq Q_i$. Assume the Type III $\times$ Type III level condition: $Q_1 Q_2 \leq (Xy)^{2/3-\epsilon}$.

For integers $q, r \geq 2$ and residue classes $a \pmod q$, $b \pmod r$ with $(a(a+4), q) = 1$ and $(b(b+4), r) = 1$, let $l := \mathrm{lcm}(q, r)$. If a and b are CRT-compatible, write $c = c(a, b)$ for the unique class mod l with $c \equiv a \pmod q, c \equiv b \pmod r$; otherwise set $\Delta(x, y; q, a; r, b; 4) = 0$. Define the short-interval discrepancy as:

$$\Delta(x, y; q, a; r, b; 4) := \sum_{\substack{x < n \leq x+y \\ n \equiv a \pmod q \\ n \equiv b \pmod r}} \Lambda(n)\Lambda(n+4) - c(l, c; 4)y$$

where $c(l, c; 4) := \left(\frac{\mathfrak{S}(4)}{\varphi(l)}\right) \cdot \mathbf{1}_{(c,l)=1} \cdot \mathbf{1}_{(c+4,l)=1} \cdot \prod_{p|l} \kappa_p(c; 4)$.Then for all $A > 0$:

(1) Almost-all short intervals: For all $x \in [X, 2X]$ outside a set of measure $\ll_A X(\log X)^{-A}$,

$$\sum_{2 \leq q \leq Q_1} \sum_{2 \leq r \leq Q_2} |\lambda_q^{(1)} \lambda_r^{(2)}| \cdot \max_{a \pmod q, b \pmod r} |\Delta(x, y; q, a; r, b; 4)| \ll_{A,\epsilon,\varpi} y(\log X)^{-A}.$$

(2) Mean-square strengthening (variance form):

$$\sum_{2 \leq q \leq Q_1} \sum_{2 \leq r \leq Q_2} |\lambda_q^{(1)} \lambda_r^{(2)}|^2 \cdot \sum_{a \pmod q} \sum_{b \pmod r} |\Delta(x, y; q, a; r, b; 4)|^2 \ll_{A,\epsilon,\varpi} y^2(\log X)^{-A}.$$

Conjecture A.74. Fix $\epsilon, \varpi > 0$ and $A > 0$. Let X be large, $x \in [X, 2X]$, and $X^{2/3+\varpi} \leq y \leq X$. For $i = 1, 2$, let $\lambda^{(i)} = (\lambda_m^{(i)})_{m \geq 2}$ be triple well-factorable sequences of levels Q_i, meaning there exist $\beta_j^{(i)} : \mathbb{N} \to \mathbb{C}$ $(j = 1, 2, 3)$ with:supp $\beta_j^{(i)} \subset (Q_{i,j}, 2Q_{i,j}]$, $\|\beta_j^{(i)}\|_\infty \ll (\log X)^{O(1)}$,$Q_{i,1}Q_{i,2}Q_{i,3} = Q_i$, and $\lambda^{(i)} = \beta_1^{(i)} * \beta_2^{(i)} * \beta_3^{(i)}$,$\lambda_m^{(i)} = 0$ unless $2 \leq m \leq Q_i$.Assume the Type III $\times$ Type III level condition: $Q_1 Q_2 \leq (Xy)^{2/3-\epsilon}$.For $q \leq Q_1, r \leq Q_2$, residue classes $a \pmod q, b \pmod r$ with $(a(a+6), q) = 1$ and $(b(b+6), r) = 1$, let $l := \mathrm{lcm}(q, r)$. Define the short-interval discrepancy in the intersection AP as:

$$\Delta(x, y; q, a; r, b; 6) := \sum_{\substack{x < n \leq x+y \\ n \equiv a \pmod q \\ n \equiv b \pmod r}} \Lambda(n)\Lambda(n+6) - c(l, c; 6)y$$

where the singular series for sexy primes ($h = 6$) is:

$$\mathfrak{S}(6) = 2 \cdot C_2 \cdot \prod_{\substack{p|6 \\ p>2}} \frac{p-1}{p-2}, \quad C_2 = \prod_{p>2} \left(1 - \frac{1}{(p-1)^2}\right).$$

(1) Almost-all short-interval mean bound (un-twisted)For all $A > 0$, for all $x \in [X, 2X]$ outside a set of measure $\ll_A X(\log X)^{-A}$:

$$\sum_{2 \leq q \leq Q_1} \sum_{2 \leq r \leq Q_2} |\lambda_q^{(1)} \lambda_r^{(2)}| \cdot \max_{a \pmod q, b \pmod r} |\Delta(x, y; q, a; r, b; 6)| \ll_{A,\epsilon,\varpi} y(\log X)^{-A}.$$

(2) 3-adic twisted equidistribution (Sexy-prime specific)Let χ_3 be the nontrivial Dirichlet character modulo 3. Then for almost all x:

$$\sum_{2\le q\le Q_1}\sum_{2\le r\le Q_2}|\lambda_q^{(1)}\lambda_r^{(2)}|\cdot\max_{a \pmod q, b \pmod r}\left|\sum_{\substack{x<n\le x+y\\ n\equiv a \pmod q\\ n\equiv b \pmod r}}\Lambda(n)\Lambda(n+6)\chi_3(n)-\chi_3(c)c(l,c;6)y\right|\ll y(\log X)^{-A}.$$

Conjecture A.75 (Uniform Split)**.** For many such (q,r), one has:

$$\sum_{\substack{x<n\le x+y\\ n\equiv a \pmod q\\ n\equiv b \pmod r}}\Lambda(n)\Lambda(n+6)(\mathbf{1}_{n\equiv 1 \pmod 3}-\mathbf{1}_{n\equiv 2 \pmod 3})=o\left(\frac{y}{(\log X)^2}\right)$$

This reflects the absence of hidden 3-adic biases for sexy prime pairs.

Setup. Fix $k\ge 2$ and an admissible k-tuple $H=\{h_1,\dots,h_k\}\subset\mathbb{Z}$ (no prime p has all residues covered by $\{h_i \pmod p\}$). Let $\epsilon,\varpi>0$ and $A>0$. Let X be large, $x\in[X,2X]$, and $X^{2/3+\varpi}\le y\le X$.

Triple well-factorable weights. For $i=1,2$ let $\lambda^{(i)}=(\lambda_m^{(i)})_{m\ge 2}$ be triple well-factorable sequences of levels Q_i: there exist $\beta_j^{(i)}:\mathbb{N}\to\mathbb{C}$ ($j=1,2,3$) with supp $\beta_j^{(i)}\subset(Q_{i,j},2Q_{i,j}]$, $\|\beta_j^{(i)}\|_\infty\ll(\log X)^{O(1)}$, $Q_{i,1}Q_{i,2}Q_{i,3}=Q_i$, and $\lambda^{(i)}=\beta_1^{(i)}*\beta_2^{(i)}*\beta_3^{(i)}$, supported on $2\le m\le Q_i$.

Type III × Type III level. Assume $Q_1Q_2\le(Xy)^{2/3-\epsilon}$.

Arithmetic progression intersection. For $2\le q\le Q_1, 2\le r\le Q_2$, and residue classes $a \pmod q, b \pmod r$ with $(\prod_i(a+h_i),q)=1$ and $(\prod_i(b+h_i),r)=1$, let $l:=\mathrm{lcm}(q,r)$. If a and b are CRT-compatible, let c denote the induced class modulo l; otherwise set all discrepancies below to 0.

Main term. Let $\mathfrak{S}(H)$ be the Hardy–Littlewood singular series $\mathfrak{S}(H)=\prod_p(1-1/p)^{-k}(1-\nu_p(H)/p)$, where $\nu_p(H)$ is the number of distinct residues in $\{h_i \pmod p\}$. Define the l-restricted local density $c(l,c;H)$ so that:

$$c(l,c;H)=\frac{\mathfrak{S}(H)}{\varphi(l)}\cdot\mathbf{1}_{c+h_i,l=1 \text{ for all } i}\cdot\prod_{p\mid l}\kappa_p(c;H)$$

where $\kappa_p(c;H)$ are standard local corrections normalized by $\frac{1}{\varphi(l)}\sum_{c \pmod l}c(l,c;H)=\frac{\mathfrak{S}(H)}{\varphi(l)}$.

Discrepancy in short intervals for the intersection AP.

$$\Delta_H(x,y;q,a;r,b):=\sum_{\substack{x<n\le x+y\\ n\equiv a \pmod q\\ n\equiv b \pmod r}}\prod_{i=1}^k\Lambda(n+h_i)-c(l,c;H)y.$$

Conjecture A.76. For all $A>0$, for all $x\in[X,2X]$ outside a set of measure $\ll_A X(\log X)^{-A}$:

$$\sum_{2\le q\le Q_1}\sum_{2\le r\le Q_2}|\lambda_q^{(1)}\lambda_r^{(2)}|\cdot\max_{a \pmod q, b \pmod r}|\Delta_H(x,y;q,a;r,b)|\ll_{A,\epsilon,\varpi,H}y(\log X)^{-A}.$$

Conjecture A.77 (Small-modulus twisted equidistribution)**.** Let $M_0(H) := \mathrm{rad}(\gcd_{i<j}(h_i - h_j))$. For any nontrivial Dirichlet character χ modulo a divisor $M|M_0(H)$, and any (q, r, a, b) as above with $M|l$, define:

$$T_\chi(x, y; q, a; r, b; H) := \sum_{\substack{x<n\le x+y \\ n\equiv a \pmod q \\ n\equiv b \pmod r}} \left(\prod_{i=1}^{k} \Lambda(n + h_i)\right) \chi(n + h_1)$$

Then, for almost all x as above:

$$\sum_{2\le q\le Q_1} \sum_{2\le r\le Q_2} |\lambda_q^{(1)}\lambda_r^{(2)}| \cdot \max_{a \pmod q, b \pmod r} |T_\chi(x, y; q, a; r, b; H) - \chi(c+h_1)c(l, c; H)y| \ll y(\log X)^{-A}.$$

Setup. Fix a primitive GL(2) newform f of level N (holomorphic or Maass), with nebentypus ψ, and let $\lambda_f(n)$ denote its normalized Hecke eigenvalues ($\lambda_f(1) = 1$, multiplicative, satisfying Ramanujan–Petersson on average). Fix a nonzero integer shift $h \ne 0$, and parameters $\epsilon, \varpi > 0, A > 0$. Let X be large, $x \in [X, 2X]$, and $X^{2/3+\varpi} \le y \le X$.

- Triple well-factorable weights: For $i = 1, 2$ let $\lambda^{(i)} = (\lambda_m^{(i)})_{m\ge2}$ be triple well-factorable sequences of levels Q_i: there exist $\beta_j^{(i)} : \mathbb{N} \to \mathbb{C}$ $(j = 1, 2, 3)$ withsupp $\beta_j^{(i)} \subset (Q_{i,j}, 2Q_{i,j}]$, $\|\beta_j^{(i)}\|_\infty \ll (\log X)^{O(1)}$,$Q_{i,1}Q_{i,2}Q_{i,3} = Q_i$, and $\lambda^{(i)} = \beta_1^{(i)} * \beta_2^{(i)} * \beta_3^{(i)}$ (Dirichlet convolution),$\lambda_m^{(i)} = 0$ unless $2 \le m \le Q_i$.
- Type III × Type III product-level condition: Assume $Q_1Q_2 \le (Xy)^{2/3-\epsilon}$.
- Intersection AP and discrepancy: For $2 \le q \le Q_1$, $2 \le r \le Q_2$ and residue classes $a \pmod q$, $b \pmod r$, let $l := \mathrm{lcm}(q, r)$. If a and b are CRT-compatible, write $c \in \mathbb{Z}/l\mathbb{Z}$ for the induced class (else set the discrepancy below to 0). Define the short-interval AP-restricted shifted-convolution discrepancy by

$$\Delta_{f,h}(x, y; q, a; r, b) := \sum_{\substack{x<n\le x+y \\ n\equiv a \pmod q \\ n\equiv b \pmod r}} \lambda_f(n)\lambda_f(n + h).$$

  (For $h \ne 0$ the expected main term is 0; for $h = 0$ one expects a positive main term proportional to $y/\phi(l)$ determined by local Rankin–Selberg factors at primes dividing lN.)

Conjecture A.78. For all $A > 0$, for all $x \in [X, 2X]$ outside a set of measure $\ll_A X(\log X)^{-A}$,

$$\sum_{2\le q\le Q_1} \sum_{2\le r\le Q_2} |\lambda_q^{(1)}\lambda_r^{(2)}| \cdot \max_{a \pmod q, b \pmod r} |\Delta_{f,h}(x, y; q, a; r, b)| \ll_{A,\epsilon,\varpi,f,h} y(\log X)^{-A}.$$

Variance strengthening (L^2 form). For the same x and y,

$$\sum_{2\le q\le Q_1} \sum_{2\le r\le Q_2} |\lambda_q^{(1)}\lambda_r^{(2)}|^2 \cdot \sum_{a \pmod q} \sum_{b \pmod r} |\Delta_{f,h}(x, y; q, a; r, b)|^2 \ll_{A,\epsilon,\varpi,f,h} y^2(\log X)^{-A}.$$

Small-modulus twisted equidistribution. Let $M \mid \mathrm{rad}(h)$ and χ be any nontrivial Dirichlet character modulo M, inflated to modulus l by composition with reduction mod M. Whenever $M \mid l$, define

$$T_{f,h,\chi}(x, y; q, a; r, b) := \sum_{\substack{x<n\le x+y \\ n\equiv a \pmod q \\ n\equiv b \pmod r}} \lambda_f(n)\lambda_f(n + h)\chi(n).$$

Then, for almost all x as above,

$$\sum_{2\le q\le Q_1}\sum_{2\le r\le Q_2}|\lambda_q^{(1)}\lambda_r^{(2)}|\cdot\max_{a\pmod q,\, b\pmod r}|T_{f,h,\chi}(x,y;q,a;r,b)|\ll_{A,\epsilon,\varpi,f,h} y(\log X)^{-A}.$$

A.2.7. *Matrix.* Total number of inequalities: 34.
Total incorrect inequalities: 1.
Rate of correctness: 97.16%.

Conversations	Total Conjectures	Incorrect Conjectures
Conversation 1	10	0
Conversation 2	1	0
Conversation 3	2	0
Conversation 4	9	1
Conversation 5	3	0
Conversation 6	9	0